Area Under Curves

DISCLAIMER

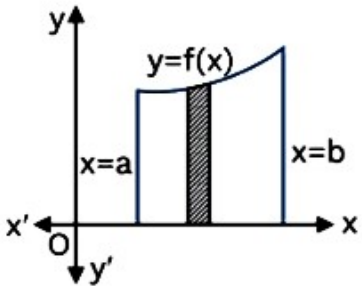
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Area Under Curves

DIFFERENT CASES OF BOUNDED AREA

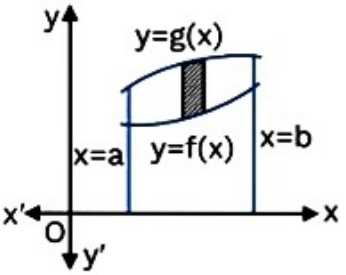
1. The area bounded by the continuous curve $y = f(x)$ $(f(x) > 0)$, the axis of x and the ordinates $x = a$ and $x = b$ (where $b > a$) is given by

$$A = \int_a^b f(x)dx = \int_a^b ydx$$



2. The area bounded by the straight line $x = a$, $x = b$ $(a < b)$ and the curves $y = f(x)$ and $y = g(x)$, provided $f(x) < g(x)$ (where $a \le x \le b$), is given by

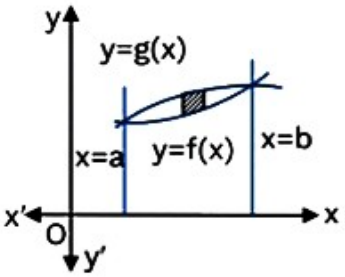
$$A = \int_a^b [g(x) - f(x)]dx$$



3. When two curve $y = f(x)$ and $y = g(x)$ intersects, the bounded area is

$$A = \int_a^b [g(x) - f(x)]dx \text{ ; where } a < b$$

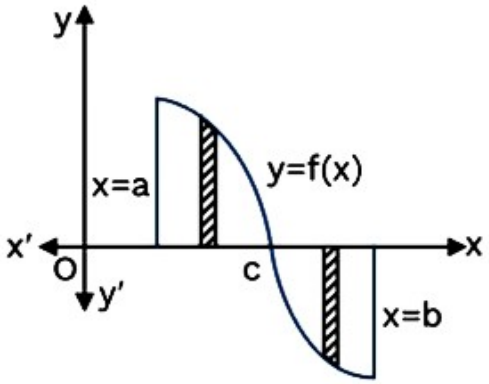
Where a and b are the roots of the equation $f(x) = g(x)$.



4. If some part of a curve lies below the x-axis, then its area becomes negative but area cannot be negative. Therefore, we take its modulus.
It the curve crosses the x-axis at c, then the area bounded by the curve $y=f(x)$ and ordinates $x = a$ and $x = b$.
(where $b > a$) is given by

$$A = \left|\int_a^c f(x)dx\right| + \left|\int_c^b f(x)dx\right|$$

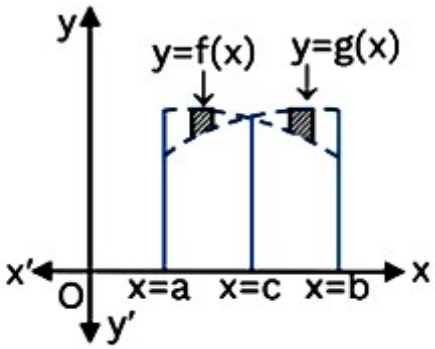
$$A = \int_a^c f(x)dx - \int_c^b f(x)dx$$



5. The area bounded by $y = f(x)$ and $y = g(x)$ (where $a \le x \le b$), when they intersect at $x = c \in (a, b)$ is given by

$$A = \int_a^b |f(x) - g(x)|\,dx$$

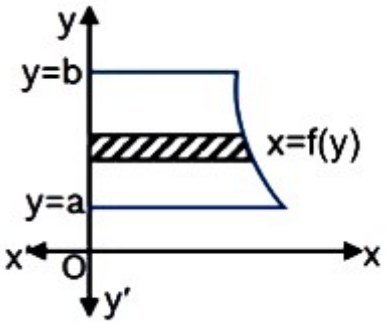
Or $$\int_a^c [f(x) - g(x)]\,dx + \int_c^b [g(x) - f(x)]\,dx$$



DIFFERENT CASES OF BOUNDED AREA

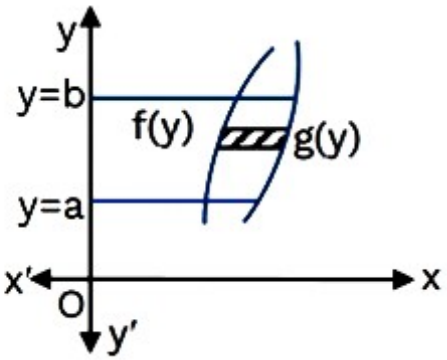
1. The area bounded by the continuous curve $x = f(y)$, the axis of y and the abscissa $y = a$ and $y = b$ (where $b > a$) is given by

$$A = \int_a^b f(y)dy = \int_a^b x\,dy$$



2. The area bounded by the straight line $y = a$, $y = b$ $(a < b)$ and the curves $x = f(y)$ and $x = g(y)$, provided $f(y) < g(y)$ (where $a \le y \le b$), is given by

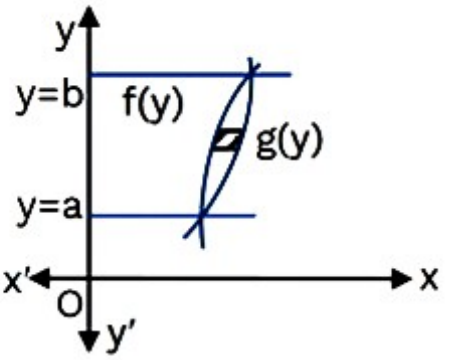
$$A = \int_a^b [g(y) - f(y)]dy$$



3. When two curves $x = f(y)$ and $x = g(y)$ intersect, the bounded area is

$$A = \int_a^b [g(y) - f(y)]dy; \quad \text{Where } a < b$$

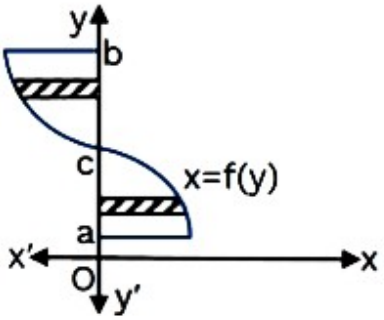
Where a and b are the roots of the equation $f(y) = g(y)$



4. If some part of a curve lies left to y-axis, then its area becomes negative but area cannot be negative. Therefore, we take its modulus.
If the curves crosses the y-axis at c, then the area bounded by the curve $x = f(y)$ and abscissae $y = a$ and $y = b$
(where $b > a$) is given by

$$A = \left|\int_a^c f(y)dy\right| + \left|\int_c^b f(y)dy\right|$$

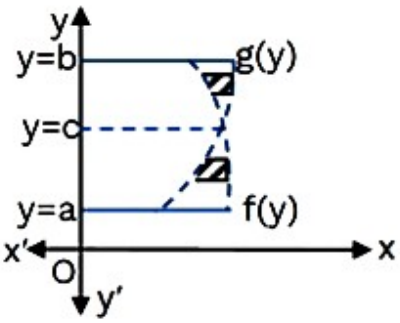
$$A = \int_a^c f(y)dy - \int_c^b f(y)dy$$



5. The area bounded by $x = f(y)$ and $x = g(y)$ (where $a \le y \le b$), when they intersect at $y = c \in (a, b)$ is given by

$$A = \int_a^b |f(y) - g(y)|\,dy$$

or $$A = \int_a^c (f(y) - g(y))dy + \int_c^b (g(y) - f(y))dy$$



ILLUSTRATIONS :

Q.1 **Find the area bounded by the parabola $y = x^2 + 1$ and the straight line $x + y = 3$.**

Sol. The two curves meet at points where $3 - x = x^2 + 1$ i.e. $x^2 + x - 2 = 0$

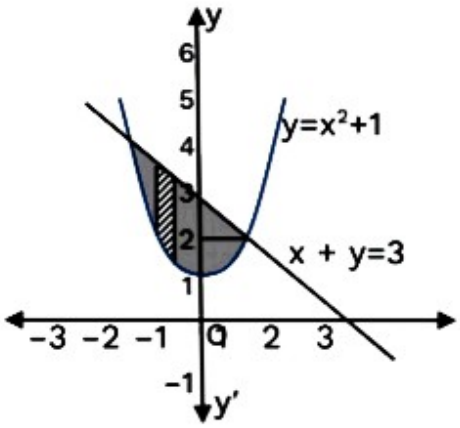
$\Rightarrow$ $(x + 2)(x - 1) = 0 \Rightarrow x = -2, 1$

$\therefore$ required area $= A = \int_{-2}^{1} \left[(3 - x) - (x^2 + 1)\right]dx$

$$A = \int_{-2}^{1} \left(2 - x - x^2\right)dx$$

$$= \left[2x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2}^{1}$$

$$= \left(2 - \frac{1}{2} - \frac{1}{3}\right) - \left(-4 - \frac{4}{2} + \frac{8}{3}\right) = \frac{9}{2} \text{ sq. units.}$$



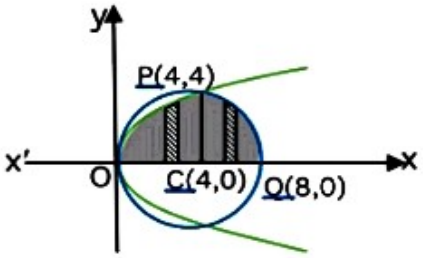
Q.2 **Find the area, lying above x-axis and included between the circle $x^2 + y^2 = 8x$ and the parabola $y^2 = 4x$.**

Sol. Solving the curves, we get $x^2 + 4x = 8x \Rightarrow x = 0, 4$.

Required area $= \int_0^4 y_{parabola}\,dx + \int_4^8 y_{circle}\,dx$

Circle is $(x-4)^2 + y^2 = 4^2$

Area of circle in 1st quadrant $= \frac{1}{4}\pi 4^2 = 4\pi$



$$A = 2\int_0^4 \sqrt{x}\,dx + 4\pi$$

$$= \frac{4}{3}\left[x^{3/2}\right]_0^4 + 4\pi = \frac{4}{3} \times 4\sqrt{4} + 4\pi \text{ sq. units}$$

$$= \frac{32}{3} + 4\pi \text{ sq. units.}$$

Q.3 **Find the area bounded by the curve $y = (x-1)(x-2)(x-3)$ lying between the ordinates $x = 0$ and $x = 3$.**

Sol. $y = (x-1)(x-2)(x-3)$

The curves will intersect the x-axis, when $y = 0$

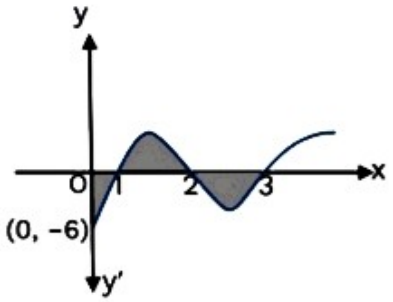
$\Rightarrow (x-1)(x-2)(x-3) = 0$

$\Rightarrow x = 1, 2, 3$

And the curve intersects the y-axis.

When $x = 0 \Rightarrow y = -6$

Thus, the graph of the given function for $0 \le x \le 3$ is as shown in figure.



Hence, the required area A = shaded area

$$= \left|\int_0^1 y\,dx\right| + \left|\int_1^2 y\,dx\right| + \left|\int_2^3 y\,dx\right|$$

Since $\int y\,dx = \int (x-1)(x-2)(x-3)\,dx$

$$= \int (x^3 - 6x^2 + 11x - 6)\,dx$$

$$= \frac{x^4}{4} - 2x^3 + \frac{11x^2}{2} - 6x \qquad \ldots(1)$$

$\therefore$ from equation (1)

$$A = \left|\left[\frac{x^4}{4} - 2x^3 + \frac{11x^2}{2} - 6x\right]_0^1\right| + \left|\left[\frac{x^4}{4} - 2x^3 + \frac{11x^2}{2} - 6x\right]_1^2\right| + \left|\left[\frac{x^4}{4} - 2x^3 + \frac{11x^2}{2} - 6x\right]_2^3\right|$$

$$= \left|\frac{-9}{4}\right| + \left(\frac{1}{4}\right) + \left|\frac{-1}{4}\right| = \frac{11}{4} \text{ sq. units.}$$

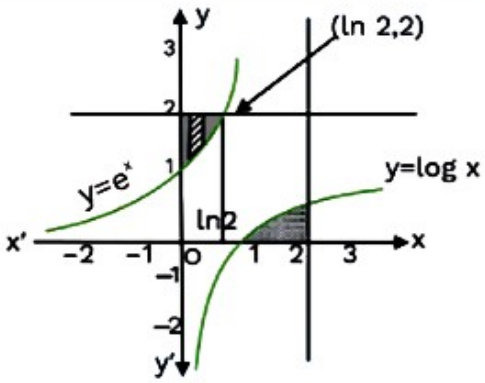
Q.4 **Consider the region formed by the lines x = 0, y = 0, x = 2, y = 2. Area enclosed by the curves $y = e^x$ and y = ln x, within this region is being removed. Then, find the area of the remaining region**

Sol. Required area = shaded region

$$= 2\int_0^{\ln 2} (2 - e^x)dx \quad \text{(By symmetry)}$$

$$= 2\left[2x - e^x\right]_0^{\ln 2}$$

$$= 2(2\ln 2 - 1) \text{ sq. units}$$



Q.5 **Find the area bounded by the curves y = sin x and y = cos x between two consecutive points of their intersection.**

Sol. Two consecutive points of intersection of y = sin x and y = cosx can be taken as

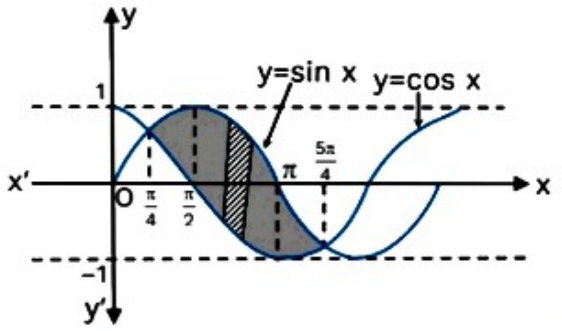
$$x = \frac{\pi}{4} \text{ and } x = \frac{5\pi}{4}$$

$$\therefore \text{ Required area} = \int_{\pi/4}^{5\pi/4} (\sin x - \cos x)dx$$

$$= \left[-\cos x - \sin x\right]_{\pi/4}^{5\pi/4}$$

$$= \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}}$$

$$= 2\sqrt{2} \text{ sq. units.}$$



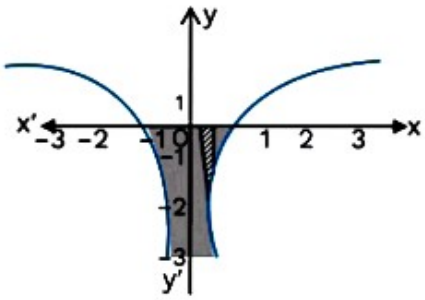
Q.6 **Find the area bounded by**

(i) $\log_e |x|$ and y = 0 **(ii) $y = |\log_e |x||$ and y = 0**

Sol. **(i)** $y = \log_e |x|$ and $y = 0$

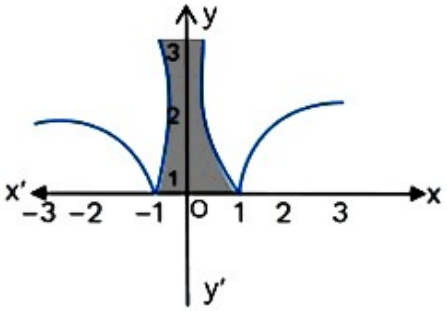
From the figure, required area = area of the shaded region

$$= 2\left|\int_0^1 (\log_e x)dx\right| = 2\left|(x\log_e x - x)_0^1\right| = 2 \text{ sq. units}$$



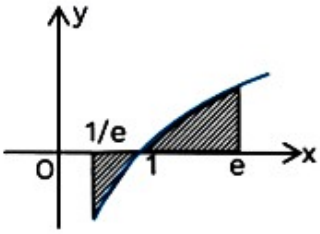
(ii) $y = |\log_e| x||$ and $y = 0$

From the figure, required area
= area of the shaded region
= 1 + 1 = 2 sq. units



Q.7 **Find the area included by the curve $y = \ln x$, x-axis and the two ordinates at $x = \frac{1}{e}$ and $x = e$.**

Sol.
$$A = \left|\int_{1/e}^{1} \ln x dx\right| + \int_1^e \ln x dx$$
$$= \left|\left[x(\ln x - 1)\right]_{1/e}^{1}\right| + \left[x(\ln x - 1)\right]_1^e$$
$$= 2 - \frac{2}{e}$$

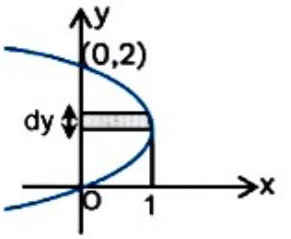


Q.8 **Find the area included by the curve $x = 2y - y^2$ and the y-axis**

Sol. $x = 2y - y^2$

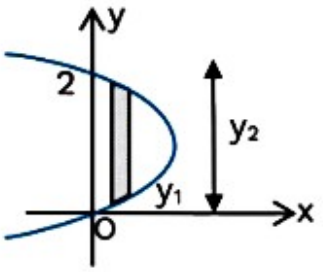
$\frac{dx}{dy} = 2 - 2y = 0 \Rightarrow y = 1$ curve bends at $y = 1$

$$A = \int_0^2 x dy = \int_0^2 (2y - y^2)dy = y^2 - \frac{y^3}{3}\Big]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$$



Alternative method:

This can also be done by taking vertical strip



$$y^2 - 2y + x = 0$$

$$y = \frac{2 \pm \sqrt{4 - 4x}}{2}$$

$$y = 1 + \sqrt{1 - x} \quad (y_2)$$

$$y = 1 - \sqrt{1 - x} \quad (y_1)$$

$$A = \int_0^1 (y_2 - y_1)\,dx = \int_0^1 2\left(\sqrt{1 - x}\right)dx = \frac{2(1-x)^{3/2}}{\left(\frac{-3}{2}\right)}\Bigg]_0^1 = \frac{-4}{3}(0 - 1) = \frac{4}{3}$$

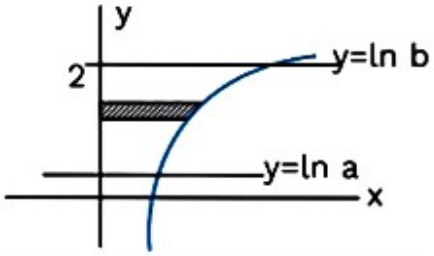
Q.9 **For $b > a > 1$, the area enclosed by the curve $y = \ln x$, y axis and the straight lines $y = \ln a$ and $y = \ln b$ is**

(A) $b - a$ **(B) $b(\ln b - 1) - a(\ln a - 1)$**

(C) $(\ln a)(b - a)$ **(D) $(\ln b)(\ln a)$**

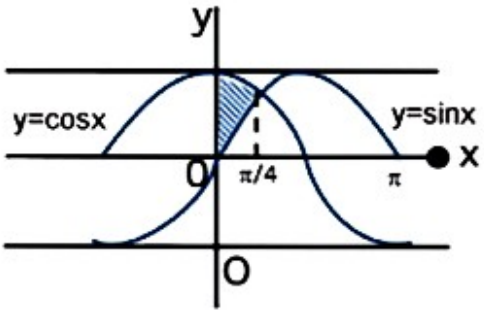
Sol. **(A)**

$$\text{Required area} = \int_{\ln a}^{\ln b} e^y dy = \left[e^y\right]_{\ln a}^{\ln b} = (b - a)$$



Q.10 **Find the area enclosed between $y = \sin x$; $y = \cos x$ and y-axis in the 1st quadrant.**

Sol. $$A = \int_0^{\pi/4} (\cos x - \sin x)\,dx = \left[\sin x + \cos x\right]_0^{\pi/4} = \sqrt{2} - 1$$



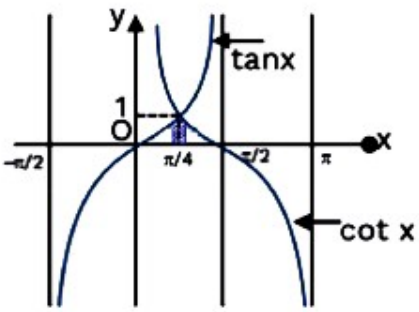
Q.11 **Find the area enclosed by y = tan x; y = cot x and x-axis in 1st quadrant.**

Sol.

$$A = \int_0^{\pi/4} \tan x\,dx + \int_{\pi/4}^{\pi/2} \cot x\,dx$$

$$A = 2\int_0^{\pi/4} \tan x\,dx = 2\left[\ln|\sec x|\right]_0^{\frac{\pi}{4}}$$

$$= 2\ln\sqrt{2} = \ln 2$$

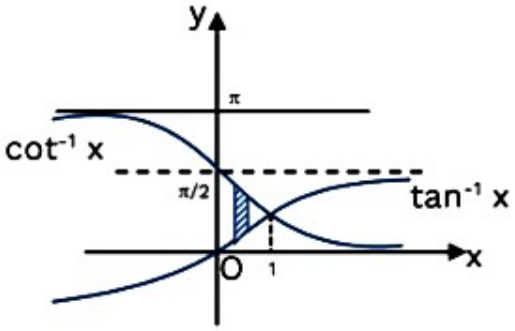


Q.12 **Compute the area enclosed between $y = \tan^{-1}x$; $y = \cot^{-1}x$ and y-axis.**

Sol.

$$A = \int_0^1 (\cot^{-1} x - \tan^{-1} x)dx$$

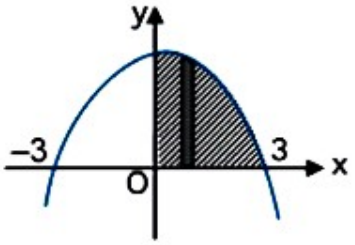
$$A = \int_0^{\pi/4} (\tan y)dy + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cot y)dy = \ln 2$$



Q.13 **Area enclosed by $y = 9 - x^2$ and coordinates axes.**

Sol.

$$A = \int_0^3 (9 - x^2)dx = 18$$

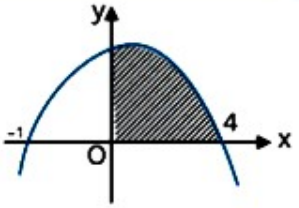


Q.14 **Compute the larger area bounded by $y = 4 + 3x - x^2$ and the coordinates axes.**

Sol.

$$A = \int_0^4 y\,dx = \int_0^4 (4 + 3x - x^2)dx$$

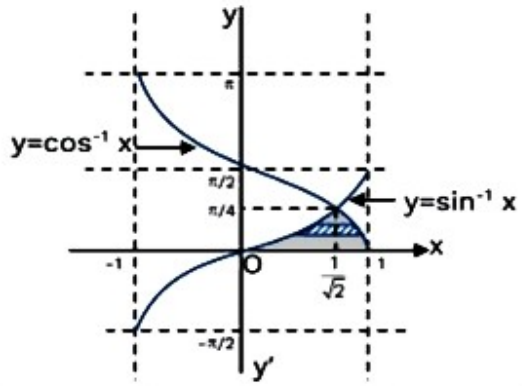
$$= \left[4x + \frac{3}{2}x^2 - \frac{1}{3}x^3\right]_0^4 = \frac{56}{3}$$



Q.15 **Find the area bounded by $y = \sin^{-1}x$, $y = \cos^{-1}x$ and x-axis.**

Sol. $y = \sin^{-1}x$, $y = \cos^{-1}x$ and the x-axis if vertical strip is used, we get

$$A = \int_0^{1/\sqrt{2}} \sin^{-1} x\,dx + \int_{1/\sqrt{2}}^{1} \cos^{-1} x\,dx$$



If horizontal strip is used, then

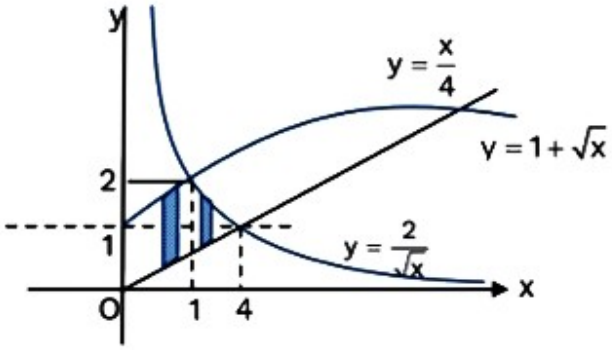
$$A = \int_0^{\pi/4} (\cos y - \sin y)\,dy$$

$$= \left[\sin y + \cos y\right]_0^{\pi/4}$$

$$= \left[\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1\right] = \sqrt{2} - 1$$

Q.16 **Find the area of the region in the 1st quadrant bounded on the left by the y-axis, below by the line $y = \frac{x}{4}$, above left by the curve $y = 1 + \sqrt{x}$ and above right by the curve $y = \frac{2}{\sqrt{x}}$.**

Sol. Required area



$$= \int_0^1 \left(1 + \sqrt{x} - \frac{x}{4}\right)dx + \int_1^4 \left(\frac{2}{\sqrt{x}} - \frac{x}{4}\right)dx$$

$$= \left(x + \frac{2}{3}x^{3/2} - \frac{x^2}{8}\right)_0^1 + \left(4\sqrt{x} - \frac{x^2}{8}\right)_1^4$$

$$= \left(1 + \frac{2}{3} - \frac{1}{8}\right) + \left(4 - \frac{15}{8}\right) = \frac{5}{3} + 2 = \frac{11}{3} \text{ sq. units.}$$

Q.17 **Find the area of the figure bounded by the parabolas $x = -2y^2, x = 1 - 3y^2$.**

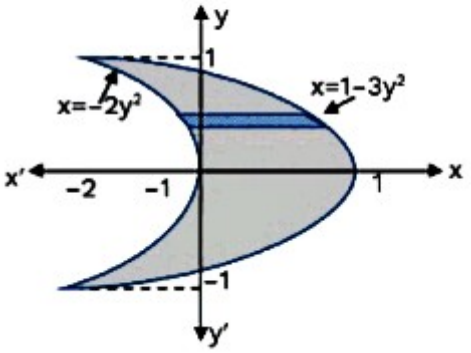
Sol. Solving the equation $x = -2y^2, x = 1 - 3y^2$ we find that ordinates of the point of intersection of the two curves as $y_1 = -1, y_2 = 1$. The points are $(-2, -1)$ and $(-2, 1)$.

The required area (using horizontal strip)

$$A = 2\int_0^1 (x_1 - x_2)dy$$

$$= 2\int_0^1 \left[\left(1-3y^2\right)-\left(-2y^2\right)\right]dy$$

$$A = 2\int_0^1 \left(1-y^2\right)dy = 2\left[y - \frac{y^3}{3}\right]_0^1 = \frac{4}{3}$$



Q.18 **Find the area bounded by the x-axis, part of the curve $y = 1 + \frac{8}{x^2}$ and the ordinates x = 2 and x = 4.**

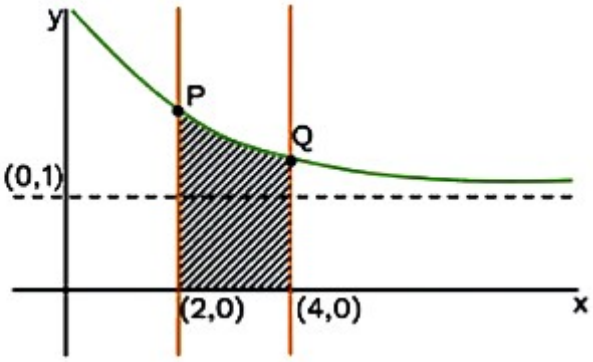
Sol. $y = 1 + \frac{8}{x^2}$ meet $x = 2$ and $x = 4$

at (2, 3) and $\left(4, \frac{3}{2}\right)$ respectively.

$$\text{Area} = \int_2^4 \left(1 + \frac{8}{x^2}\right)dx$$

$$= \left[x - \frac{8}{x}\right]_2^4$$

$$= (4-2) - (2-4) = 4.$$



Q.19 **Show that the area enclosed by the line segments $|x| + |y| = 1$ is 2.**

Sol.

Case-I : $x \geq 0, y \geq 0 \rightarrow 1^{st}$ Quadrant

$\Rightarrow x + y = 1$

Case-II : $x \geq 0, y \leq 0 \rightarrow 4^{th}$ Quadrant

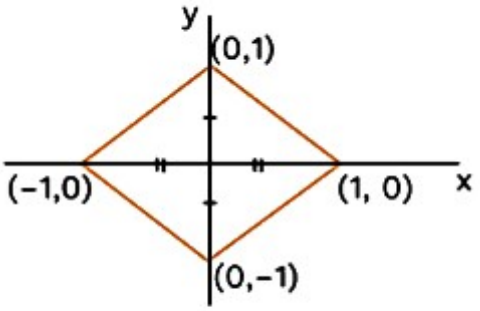
$\Rightarrow x - y = 1$

Case-III : $x \leq 0, y \geq 0 \rightarrow 2^{nd}$ Quadrant

$\Rightarrow -x + y = 1.$

Case-IV : $x < 0, y < 0 \rightarrow 3^{rd}$ Quadrant

$\Rightarrow x + y = -1$



$$\text{Area} = \frac{1}{2}\times 2\times 2 \qquad \left(A = \frac{1}{2}d_1d_2 \text{ in Rhombus}\right)$$

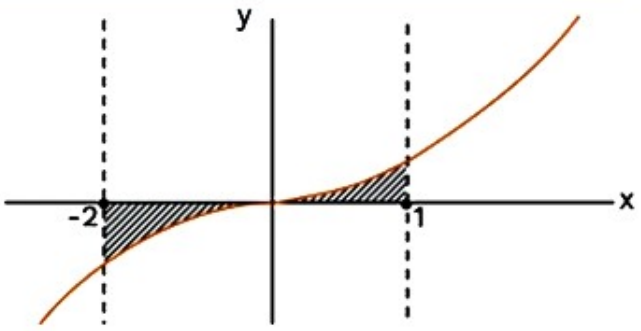
$$= 2$$

Hence proved.

Q.20 **The area bounded by curve $y = x^3$, the x-axis and the ordinate at $x = -2$ and $x = 1$ is**

(A) $\frac{9}{2}$ **(B)** $\frac{15}{2}$ **(C)** $\frac{15}{4}$ **(D)** $\frac{17}{4}$

Sol. **(D)**



$$\text{Area} = \left|\int_{-2}^{0} x^3dx\right| + \int_{0}^{1} x^3dx$$

$$= \frac{-x^4}{4}\Big]_{-2}^{0} + \frac{x^4}{4}\Big]_{0}^{1}$$

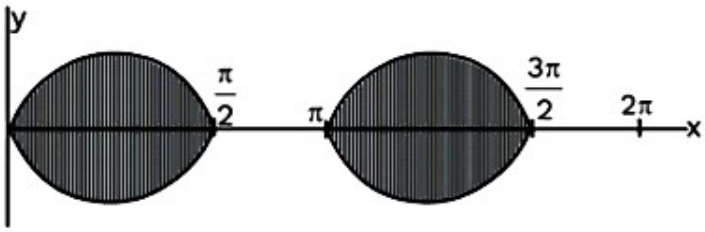
$$= 0 - (-4) + \frac{1}{4} - 0$$

$$= \frac{17}{4}$$

Q.21 **The area enclosed by the curve $|y| = \sin 2x$, when $x \in [0, 2\pi]$ is**

(A) 1 **(B) 2** **(C) 3** **(D) 4**

Sol. **(D)**



$|y| = \sin 2x \Rightarrow \sin 2x \geq 0$

$\Rightarrow 2x \in [0, \pi] \cup [2\pi, 3\pi] \cup \{4\pi\}$

$\Rightarrow x \in \left[0, \frac{\pi}{2}\right] \cup \left[\pi, \frac{3\pi}{2}\right] \cup \{2\pi\}$

Also $|y| = \sin 2x$ represents two curve, $y = \sin 2x$ and $y = -\sin 2x$, both periodic function with period π.

From the graph, using symmetry, we can say that area enclosed is equal to :

$$A = 4\int_{0}^{\frac{\pi}{2}} \sin 2x\,dx = -2\cos 2x\Big]_{0}^{\frac{\pi}{2}} = -2(-1-1) = 4$$

Q. **The area bounded by the curve $y = 2x - x^2$ and the straight line $y = -x$ is given by**

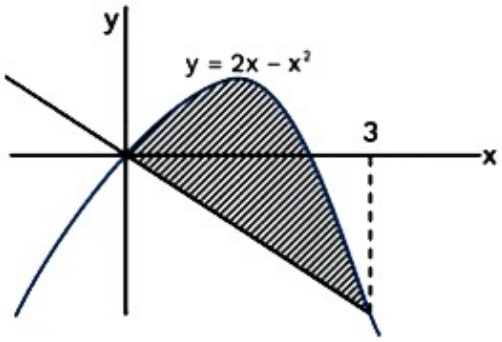
(A) $\frac{9}{2}$ **(B)** $\frac{43}{6}$ **(C)** $\frac{35}{6}$ **(D)** $\frac{1}{3}$

Sol. **(A)**

Points of intersection: $y = 2x - x^2 = -x$

$\Rightarrow x^2 - 3x = 0$

$\Rightarrow x = 0$ or 3.



$$\Rightarrow \text{Area} = \int_0^3 \left[\left(2x - x^2\right) - \left(-x\right)\right] dx$$

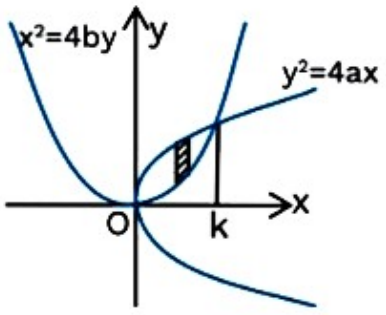
$$= \int_0^3 \left(3x - x^2\right) dx$$

$$= \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3$$

$$= \frac{27}{2} - \frac{27}{3} = \frac{9}{2}.$$

STANDARD AREAS TO BE REMEMBERED:

(1) Area bounded by the curve $y^2 = 4ax$; $x^2 = 4by$ is equal to $\left|\frac{16ab}{3}\right|$



At point of intersection

$$\left(\frac{x^2}{4b}\right)^2 = 4ax \Rightarrow x^4 = 64ab^2x$$

$\Rightarrow \quad x = 0, \ (64\ ab^2)^{1/3}$

Let $k = 4(ab^2)^{1/3}$

$$A = \int_0^k \left(2\sqrt{a}\sqrt{x} - \frac{x^2}{4b}\right) dx$$

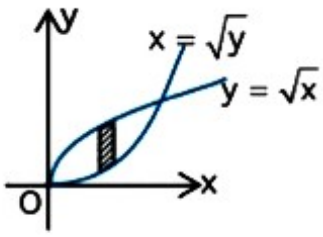
$$= \left[2\sqrt{a}\,\frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^3}{12b}\right]_0^k = \frac{4\sqrt{a}}{3}k^{\frac{3}{2}} - \frac{k^3}{12b} = \frac{4}{3}\sqrt{a}\,8\left(ab^2\right)^{\frac{1}{2}} - \frac{64(ab^2)}{12b}$$

$$= \frac{32}{3}ab - \frac{16}{3}ab = \frac{16ab}{3}$$

ILLUSTRATION:

Q. **Find the area bounded by the curve $y = \sqrt{x}; x = \sqrt{y}$**

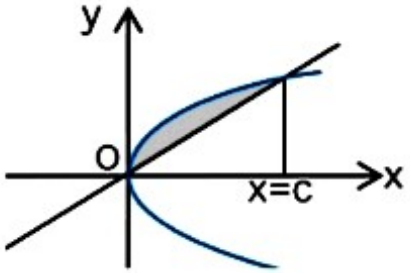
Sol. $a = \frac{1}{4}; b = \frac{1}{4}$



$$\text{Required area} = \frac{16ab}{3} = \frac{16 \cdot \frac{1}{4} \cdot \frac{1}{4}}{3}$$

$$\text{Area} = \frac{1}{3}$$

(2) Area bounded by the parabola $y^2 = 4ax$ and $y = mx$ is equal to $\left|\frac{8a^2}{3m^3}\right|$:



$y^2 = 4ax$ and $y = mx$

At point of intersection

$$m^2x^2 = 4ax \quad \Rightarrow \quad x = 0, \frac{4a}{m^2}$$

$$\text{Area} = \int_0^c \left(2\sqrt{a}\sqrt{x} - mx\right)dx \quad \text{where } c = \frac{4a}{m^2}$$

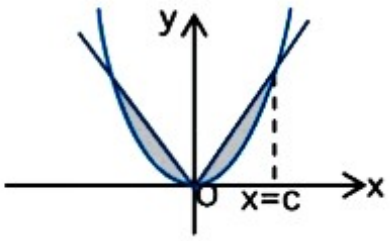
$$= \left(2\sqrt{a}\,\frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{mx^2}{2}\right)_0^c = \frac{4\sqrt{a}}{3}c^{\frac{3}{2}} - \frac{mc^2}{2}$$

$$= \frac{4\sqrt{a}}{3} \cdot \frac{8a\sqrt{a}}{m^3} - \frac{m}{2} \cdot \frac{16a^2}{m^4} = \frac{32a^2}{3m^3} - \frac{8a^2}{m^3} = \frac{8a^2}{3m^3}$$

ILLUSTRATION :

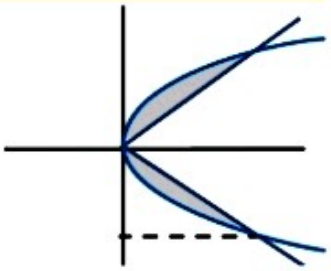
Q. **Find the area bounded by the curves $x^2 = y$; $y = |x|$.**

Sol. $$\text{Area} = 2\left(\frac{8a^2}{3m^3}\right) = 16\frac{\left(\frac{1}{4}\right)^2}{3(1)^3} = \frac{1}{3}$$

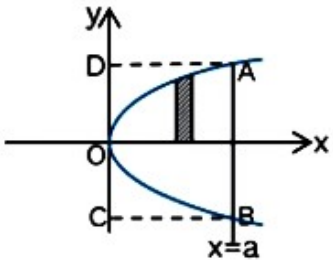


Q. **Find the area bounded by the curves $y^2 = x$; $x = |y|$**

$$\text{Area} = 2\left(\frac{8a^2}{3m^3}\right) = \frac{16}{3}\cdot\frac{\left(\frac{1}{4}\right)^2}{(1)^3} = \frac{1}{3}$$



(3) Area enclosed by $y^2 = 4ax$ and its double ordinates at $x = a$ is equal to $\frac{2}{3}$(ar□ABCD) :



(chord perpendicular to the axis of symmetry)
Required area = OABO

$$= 2\cdot\int_0^a \left(2\sqrt{ax}\right)dx = 4\sqrt{a}\left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right)_0^a$$

$$= \frac{8}{3}\sqrt{a}\cdot\left(a\sqrt{a}\right) = \frac{8a^2}{3}$$

Area of rectangle ABCD = $4a^2$

$\Rightarrow$ Area of OAOB = $\frac{2}{3}$(area □ABCD)

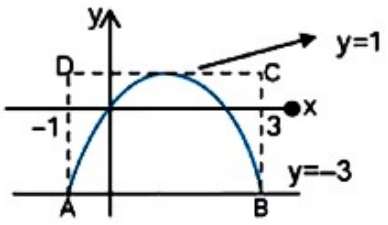
ILLUSTRATION:

Q. **Find the area bounded by the curve $y = 2x - x^2$, $y + 3 = 0$**

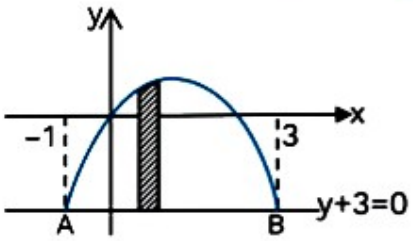
Sol. For point of intersection of $y = 2x - x^2$ and $y + 3 = 0$

Area (□ABCD) = 4 × 4 = 16

Required area = $= \frac{2}{3}\times 16 = \frac{32}{3}$



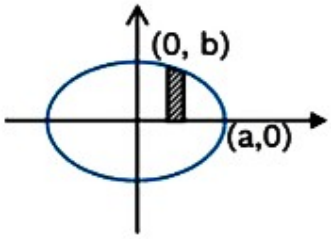
Alternative method:



By integration

$$A = \int_{-1}^{3}\left[(2x - x^2) - (-3)\right]dx = \frac{32}{3}$$

(4) Whole area of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is equal to π ab :



$$A = 4\int_{0}^{a}\left(b\sqrt{1-\frac{x^2}{a^2}}\right)dx$$

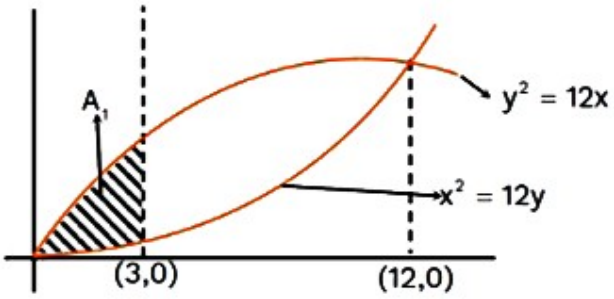
Put $x = a \sin\theta$

$$A = 4\int_{0}^{\pi/2} ab\cos^2\theta d\theta = 4ab\int_{0}^{\pi/2}\cos^2\theta d\theta$$

$$= 4ab\int_{0}^{\pi/2}\left(\frac{1+\cos 2\theta}{2}\right)d\theta = 4ab\left(\frac{\pi}{4}\right) = \pi ab$$

Q.1 **Find the ratio in which the area bounded by the curve $y^2 = 12x$ and $x^2 = 12y$ is divided by the line $x = 3$.**

Sol. Area between curves $y^2 = 4ax$ and $x^2 = 4by$ is $\frac{16ab}{3}$.



Herea a = 3 and b = 3

$$\Rightarrow \text{Area} = \frac{16(3)(3)}{3} = 48$$

$$A_1 = \int_{0}^{3}\left(\sqrt{12x} - \frac{x^2}{12}\right)dx$$

$$A_1 = \left[\frac{2\sqrt{3}x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^3}{36}\right]_0^3$$

$$= 12 - \frac{3}{4} = \frac{45}{4}$$

$$\Rightarrow \text{ ratio of division of area } = \frac{A_1}{48 - A_1}$$

$$\frac{\frac{45}{4}}{\frac{147}{4}} = \frac{45}{147} = \frac{15}{49}$$

Q.2 **Find the area of ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$**

Sol. Area of ellipse = $\pi ab = \pi(4)(3) = 12\pi$

SHIFTING OF ORIGIN :

Since area remains invariant even if the coordinates axes are shifted, hence shifting of origin in many cases proves to be very convenient in computing the areas.

ILLUSTRATION :

Q.1 **Find area enclosed between the ellipse $9x^2+4y^2-36x+8y+4=0$ and the line $3x + 2y - 10 = 0$ in the first quadrant.**

Sol. $9x^2 + 4y^2 - 36x + 8y + 4 = 0$

$\Rightarrow \quad 9(x-2)^2 + 4(y+1)^2 = 36$

$\Rightarrow \quad \frac{(x-2)^2}{2^2}+\frac{(y+1)^2}{3^2}=1$ ...(1)

Let $X = x - 2$ and $Y = y + 1$

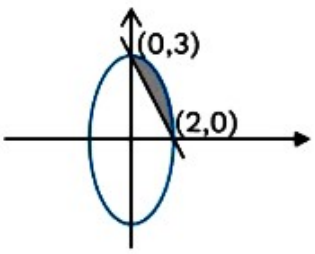
So equation of ellipse will be

$$\frac{X^2}{2^2}+\frac{Y^2}{3^2}=1$$

and equation of line $3x + 2y - 10 = 0$ ...(2)

$3(X + 2) + 2(Y - 1) - 10 = 0$

$3X + 2Y - 6 = 0$



So required area (shaded region)

$$= \frac{\pi ab}{4} - \frac{1}{2}(ab)$$

$$= \frac{\pi}{4}(2)(3) - \frac{1}{2}(2)(3) = \frac{3\pi}{2} - 3 = \frac{3(\pi-2)}{2}$$

Q.2 **Find the area enclosed between the parabola $y^2-2y+4x+5=0$ and $x^2+2x-y+2=0$**

Sol. Parabola 1 : $y^2 - 2y + 4x + 5 = 0$

$\Rightarrow (y-1)^2 = -4(x+1)$

Parabola 2 : $x^2 + 2x - y + 2 = 0$

$\Rightarrow (x+1)^2 = y-1$

$\Rightarrow$ By shifting the origin to point $(-1, 1)$,

Parabola 1 : $Y^2 = -4x$

Parabola 2 : $X^2 = Y$

$\Rightarrow$ Area enclosed $= \dfrac{16(1)\left(\frac{1}{4}\right)}{3}$ $\left(\text{area} = \dfrac{16ab}{3}, a = 1, b = \dfrac{1}{4}\right)$

$= \dfrac{4}{3}$

Q.3 **Find the area enclosed by the parabola $(y-2)^2 = x-1$ and the tangent to it at (2, 3) and x-axis.**

Sol. Shifting origin to (1, 2),

Equation of parabola becomes $y^2 = x$

Point of tangent $\equiv (1,1)$

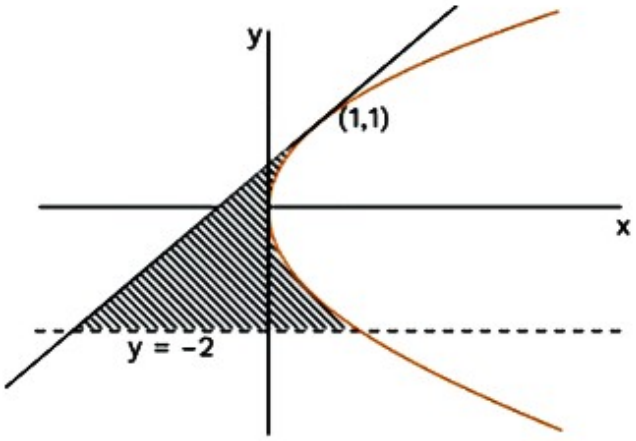
$\Rightarrow$ Tangents: $y - 1 = \dfrac{1}{2}(x-1)$

$\left(\left.\dfrac{dy}{dx}\right|_{x=1} = \dfrac{1}{2}\right)$

x-axis $\rightarrow$ $y + 2 = 0$

Graph :

From above figure, it is clear that finding the area using horizontal strip is more easy.



$\Rightarrow \text{Area} = \int_{-2}^{1} \left(y^2 - (2y-1)\right) dy$

$= \int_{-2}^{1} \left(y^2 - 2y + 1\right) dy = \left[\dfrac{y^3}{3} - y^2 + y\right]_{-2}^{1}$

$= \left(\dfrac{1}{3} - 1 + 1\right) - \left(\dfrac{-8}{3} - 4 - 2\right)$

$= \dfrac{1}{3} + \dfrac{26}{3} = 9$

Q.1 **The area enclosed between the curves $y = ax^2$ and $x = ay^2 (a > 0)$ is 1sq.unit. Then, value of a is**

(A) $\frac{1}{\sqrt{3}}$ **(B)** $\frac{1}{2}$ **(C)** 1 **(D)** $\frac{1}{3}$

Sol. **(A)**

Curves : $x^2 = \left(\frac{1}{a}\right)y$ and $y^2 = \left(\frac{1}{a}\right)x$

$\Rightarrow x^2 = 4\left(\frac{1}{4a}\right)y$ and $\Rightarrow y^2 = 4\left(\frac{1}{4a}\right)x$

Area $= \frac{16\left(\frac{1}{4a}\right)\left(\frac{1}{4a}\right)}{3}$

$= \frac{1}{3a^2} = 1$

$\Rightarrow a^2 = \frac{1}{3}$

$\Rightarrow a = \frac{1}{\sqrt{3}}$

Q.2 **Find the value of 'a' (a > 2) for which the reciprocal of the area enclosed between $y = \frac{1}{x^2}; y = \frac{1}{4(x-1)}; x = 2$ and $x = a$ is 'a' itself and for what values of $b \in (1,2)$, the area of the figure bounded by the lines x = b and x = 2 is $1 - \frac{1}{b}$.**

Sol. Point of intersection:

$\frac{1}{x^2} = \frac{1}{4(x-1)}$

$\Rightarrow x^2 = 4x - 4$

$\Rightarrow (x-2)^2 = 0$

$\Rightarrow x = 2$

$\left.\frac{dy}{dx}\right|_{x=2} = \frac{-1}{4}$ for both curves (check),

$\Rightarrow$ Both curves touches each other at x = 2.

$\Rightarrow$ Area $= \frac{1}{a}$. (between x = 2 and x = a)

$$\Rightarrow \int_2^a \left(\frac{1}{4(x-1)} - \frac{1}{x^2}\right)dx = \frac{1}{a}$$

$$\Rightarrow \frac{\ell n(x-1)}{4} + \frac{1}{x}\Bigg]_2^a = \frac{1}{a}$$

$$\Rightarrow \frac{\ell n(a-1)}{4} + \frac{1}{a} - 0 - \frac{1}{2} = \frac{1}{a}$$

$$\Rightarrow \ell n(a-1) = 2$$

$$\Rightarrow a - 1 = e^2$$

$$\Rightarrow a = 1 + e^2$$

Area enclosed in x = b and x = 2 $= 1 - \frac{1}{b}$

$$\Rightarrow \int_b^2 \left(\frac{1}{4(x-1)} - \frac{1}{x^2}\right)dx = 1 - \frac{1}{b}$$

$$\Rightarrow \frac{\ell n(x-1)}{4} + \frac{1}{x}\Bigg]_b^2 = 1 - \frac{1}{b}$$

$$\Rightarrow \frac{1}{2} - \frac{\ell n(b-1)}{4} - \frac{1}{b} = 1 - \frac{1}{b}$$

$$\Rightarrow \ell n(b-1) = -2$$

$$\Rightarrow b = 1 + \frac{1}{e^2}.$$

Q.3 **Find the area of the figure bounded by the parabola $y = ax^2 + 12x - 14$ and the straight line $y = 9x - 32$ if the tangent drawn to the parabola at the point $x = 3$ is known to make an angle $\pi - \tan^{-1} 6$ with the x-axis.**

Sol. $\tan(\pi - \tan^{-1} 6) = -6 = m = \frac{dy}{dx}$

$$\Rightarrow \frac{dy}{dx} = 2ax + 12 = -6 \text{ at } x = 3$$

$$\Rightarrow a = -3,$$

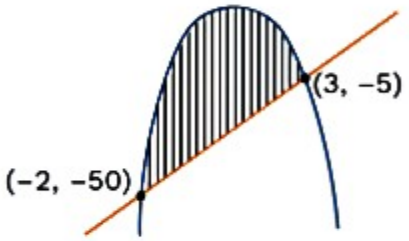
$\Rightarrow$ Point of intersection:

$-3x^2 + 12x - 14 = 9x - 32$

$\Rightarrow 3x^2 - 3x - 18 = 0$

$\Rightarrow x^2 - x - 6 = 0$

$\Rightarrow x = 3, -2$



Graph

$$\text{Area enclosed} = \int_{-2}^{3} \left(-3x^2 + 12x - 14\right) - \left(9x - 32\right) dx$$

$$= \int_{-2}^{3} \left(-3x^2 + 3x + 18\right) dx$$

$$= -x^3 + \frac{3x^2}{2} + 18x \Big]_{-2}^{3}$$

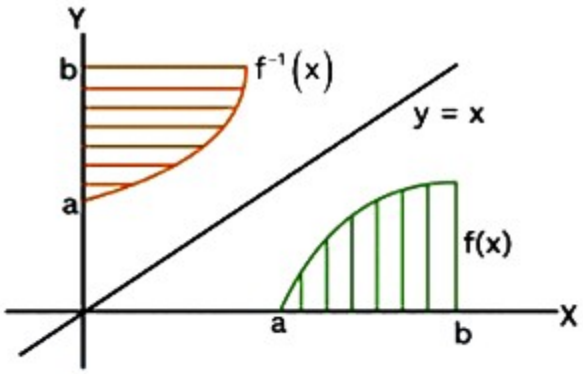
$$= \frac{81}{2} - \left(-22\right)$$

$$= \frac{125}{2}.$$

Area bounded by inverse of a function

f(x) and g(x) are inverse of each other. Since inverse functions are mirror-image of each other about y = x, we can say that,

Area bounded by f(x) and x-axis is same as area bounded by $f^{-1}(x)$, i.e.,g(x) and y-axis.



Q.1 **Find the area bounded by the inverse function of $y = f(x) = x^3 + 1$ between ordinates x = −7 and x = 2 and x-axis.**

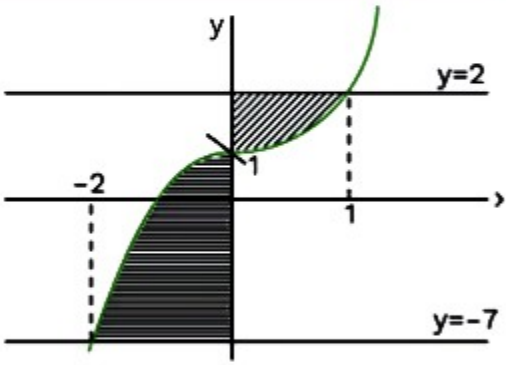
Sol. Area bounded by $f^{-1}(x)$, x-axis, x = −7 and x = 2 is same as the area bounded by f(x), y-axis , y = −7 and y = 2.

$\Rightarrow f(x) = 2 = 1 + x^3$

$\Rightarrow x = 1$

$\Rightarrow f(x) = -7 = 1 + x^3$

$\Rightarrow x = -2$



Graph

$$\text{Area} = \int_{-2}^{0}\left(x^3+1-(-7)\right)dx + \int_{0}^{1}\left(2-\left(x^3+1\right)\right)dx$$

$$= \frac{x^4}{4} + 8x\Big]_{-2}^{0} + \left(x - \frac{x^4}{4}\right)\Big]_{0}^{1}$$

$$= 0 - (4-16) + 1 - \frac{1}{4} = \frac{51}{4}$$

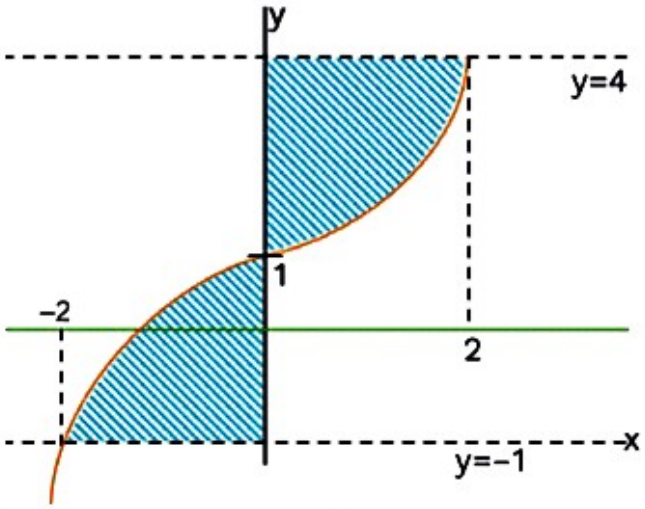
Q.2 **Find the area bounded by the curve g(x), x-axis and the ordinate at x = –1 and x = 4 where g(x) is the inverse of the function $f(x) = \frac{x^3}{24} + \frac{x^2}{8} + \frac{13x}{12} + 1$.**

Sol. Area required will be equal to the area bounded by f(x), y = –1, y = 4 and y-axis.

$$f(x) = \frac{x^3}{24} + \frac{x^2}{8} + \frac{13x}{12} + 1$$

$f(x) = -1 \Rightarrow x = -2$ (check)

$f(x) = 4 \Rightarrow x = 2$ (check)



$$\text{Area} = \int_{-2}^{0}\left[\frac{x^3}{24} + \frac{x^2}{8} + \frac{13x}{12} + 1 - (-1)\right]dx + \int_{0}^{2}\left[4 - \left(\frac{x^3}{24} + \frac{x^2}{8} + \frac{13x}{12} + 1\right)\right]dx$$

$$= \frac{x^4}{96} + \frac{x^3}{24} + \frac{13x^2}{24} + 2x\Big]_{-2}^{0} + \left(\frac{-x^4}{96} - \frac{x^3}{24} - \frac{13x^2}{24} + 3x\right)\Big]_{0}^{2}$$

$$= 2 + \frac{10}{3} = \frac{16}{3}$$

CURVE TRACING

To find the approximate shape of a curve, follow the given procedure in order

(I) **SYMMETRY:**

(a) Symmetry about x-axis

If the equation of the curve remain unchanged by replacing y by –y then the curve is symmetrical about the x-axis.

e.g., $y^2 = 4ax$

(b) Symmetry about y-axis

If the equation of the curve remain unchanged by replacing x by –x then the curve is symmetrical about the y-axis.

e.g. $x^2 = 4ay$

(c) Symmetry about both axes

If the equation of the curve remain unchanged by replacing x by –x and y by –y then the curve is symmetrical about the axis of 'x' as well as 'y'.

e.g. $x^2 + y^2 = a^2$

(d) Symmetry about the line y = x

If the equation of curve remains unchanged on interchanging 'x' and 'y', then the curve is symmetrical about the line y = x.

e.g. $x^3 + y^3 = 3xy$

(II) Find the points where the curve crosses the x-axis and the y-axis.

(III) Find $\frac{dy}{dx}$ and examine, if possible, the intervals where f(x) is increasing or decreasing and also its stationary points.

(IV) Examine y when $x \to \infty$ or $x \to -\infty$

ILLUSTRATION:

Q.1 **Draw a rough sketch of the curve, $y = \frac{x^2 + 3x + 2}{x^2 - 3x + 2}$ and find the area of the bounded region between the curve and x-axis**

Sol. $f(x) = \frac{(x+1)(x+2)}{(x-1)(x-2)}$

Graph will cut x-axis $x = -1$ and $x = -2$

It is discontinuous at $x = 1$ and $x = 2$

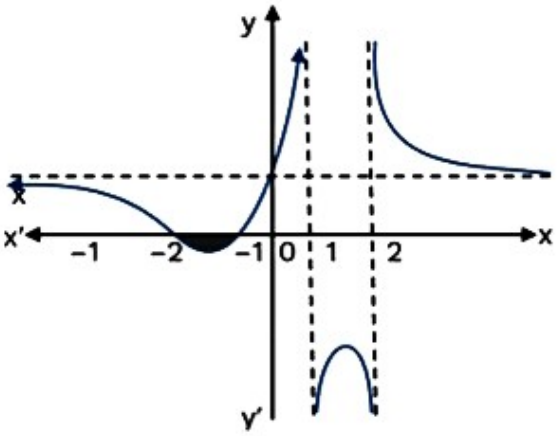
$$\lim_{x\to\pm\infty} f(x) \to 1, \lim_{x\to 1^-} f(x) \to +\infty$$

$$\lim_{x\to 1^+} f(x) \to -\infty$$

$$\lim_{x\to 2^-} f(x) \to -\infty, f(0) = 1, \lim_{x\to 2^+} f(x) = \infty;$$

Now we have to find area of the shaded region.

The required area



$$= \left|\int_{-2}^{-1} f(x)dx\right| = \left|\int_{-2}^{-1}\left(\frac{x^2+3x+2}{x^2-3x+2}\right)dx\right|$$

$$= \left|\int_{-2}^{-1}\left(1+\frac{6x}{(x-1)(x-2)}\right)dx\right|$$

$$= \left|\left[x\right]_{-2}^{-1} + 6\int_{-2}^{-1}\left(\frac{2}{x-2}-\frac{1}{x-1}\right)dx\right|$$

$$= \left|1+6\left[2\ln|x-2|-\ln|x-1|\right]_{-2}^{-1}\right|$$

$$= \left|1+6[2(\ln 3-\ln 4)-(\ln 2-\ln 3)]\right| = \left|1+6[3\ln 3-5\ln 2]\right|$$

$$= 6\ln\left(\frac{32}{27}\right)-1 \text{ sq. units}$$

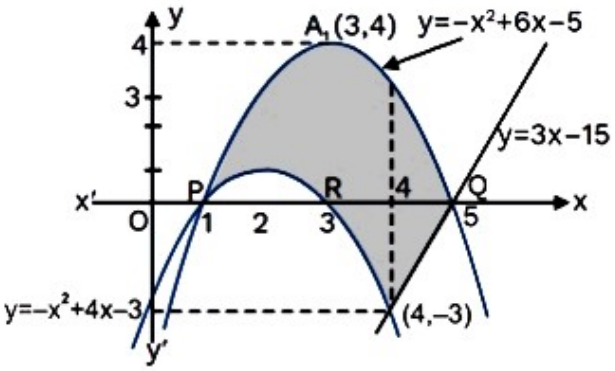
Q.2 **Find the area bounded by the curves $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$ and the straight line $y = 3x - 15$.**

Sol. The given curves are

$y = -x^2 + 6x - 5$ or $(x-3)^2 = -(y-4)$ ...(i)

Which is a parabola with vertex at A_1 (3, 4) and axis parallel to negative y-axis. It intersects the x-axis at the point P(1, 0) and Q(5, 0)

$y = -x^2 + 4x - 3$ or $(x-2)^2 = -(y-1)$ ...(ii)



which is parabola with vertex at A_2 (2,1) and axis parallel to negative y-axis.
It intersects the x-*axis* at the points P(1, 0) and R(3, 0).
And y = 3x – 15. ...(iii)
Solving, the points of intersection of (i), (ii) is (1, 0); (i), (iii) are (–2. –21) and (5, 0) and (ii), (iii) are (–3, –24) and (4, –3).
Thus, the required area is the shaded area in the diagram.

$$\text{Required area} = \left|\int_1^4 (y_1 - y_2)dx\right| + \left|\int_4^5 (y_1 - y_3)dx\right|$$

$$= \left|\int_1^4 \left[(-x^2 + 6x - 5) - (-x^2 + 4x - 3)\right]dx\right| + \left|\int_4^5 [(-x^2 + 6x - 5) - (3x - 15)]dx\right|$$

$$= \left|\int_1^4 (2x - 2)dx\right| + \left|\int_4^5 (-x^2 + 3x + 10)dx\right|$$

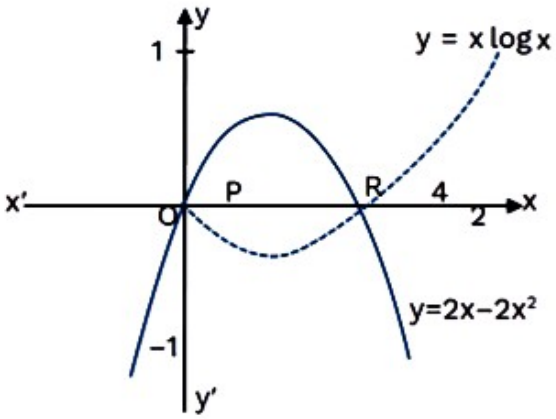
= 9 + 19/6 = 73/6 sq. units

Q. **The area of the region enclosed by the curves y = x log x and $y = 2x - 2x^2$ is**

(A) $\frac{7}{12}$ **sq. units** **(B)** $\frac{1}{2}$ **sq. units**

(C) $\frac{5}{12}$ **sq. units** **(D) None of these**

Sol. **(A)**



Curve tracing: $y = x \log_e x$
Clearly, x > 0
For 0 < x < 1, $x \log_e x < 0$
and for x > 1, $x \log_e x > 0$
Also $x \log_e x = 0 \Rightarrow x = 1$
Further, $\frac{dy}{dx} = 0 \Rightarrow 1 + \log_e x = 0$
$\Rightarrow$ x = 1/e, which is a point of minima
Required area

$$= \int_0^1 (2x - 2x^2)dx - \int_0^1 x \log x dx$$

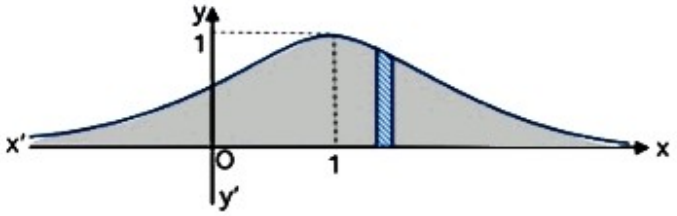
$$= \left[x^2 - \frac{2x^3}{3}\right]_0^1 - \left[\frac{x^2}{2}\log x - \frac{x^2}{4}\right]_0^1$$

$$= \left(1 - \frac{2}{3}\right) - \left[0 - \frac{1}{4} - \frac{1}{2}\lim_{x\to 0} x^2 \log x\right] = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

Q. **Area bounded by $y = \frac{1}{x^2 - 2x + 2}$ and x-axis is**

(A) 2π sq. units **(B) $\frac{\pi}{2}$ sq. units** **(C) 2 sq. units** **(D) π sq. units**

Sol. **(D)**



$$y = \frac{1}{(x-1)^2 + 1}$$

$$\text{Area} = 2\int_1^\infty \frac{1}{(x-1)^2 + 1}\,dx$$

$$= 2\left[\tan^{-1}(x-1)\right]_1^\infty = \pi \text{ sq. units}$$

Q. **Area bounded by the curve $xy^2 = a^2(a-x)$ and y-axis is**

(A) $\frac{\pi a^2}{2}$ sq. units **(B) πa^2 sq. units**

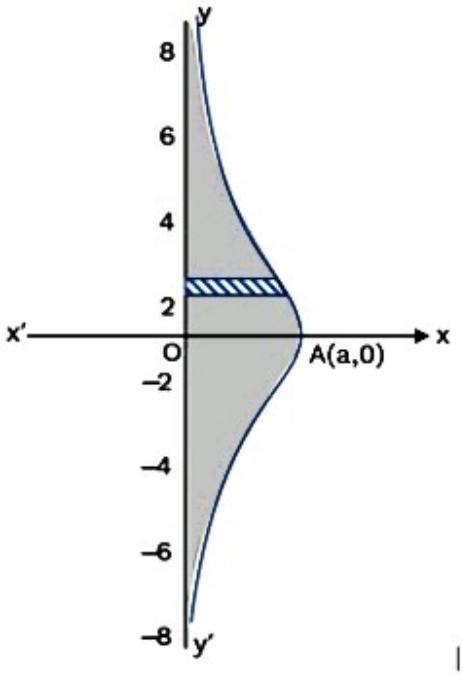
(C) $3\pi a^2$ sq. units **(D) None of these**

Sol. **(B)**

$xy^2 = a^2(a - x)$

$$\Rightarrow x = \frac{a^3}{y^2 + a^2}$$

The given curve is symmetrical about x-axis.
and meets it at (a, 0). The line $x = 0$.
i.e., y-axis is an asymptote.



$$\text{Area} = 2\int_0^\infty x\,dy = 2\int_0^\infty \frac{a^3}{y^2 + a^2}\,dy$$

$$= 2a^3 \frac{1}{a}\left[\tan^{-1}\frac{y}{a}\right]_0^\infty$$

$$= 2a^2 \frac{\pi}{2} = \pi a^2 \text{ sq. units.}$$

Q. **The area between the curve $y = 2x^4 - x^2$, the x-axis and the ordinates of the two minima of the curves is**

(A) $\frac{11}{60}$ **sq. units** **(B)** $\frac{7}{120}$ **sq. units** **(C)** $\frac{1}{30}$ **sq. units** **(D)** $\frac{7}{90}$ **sq. units**

Sol. **(B)**

The curve is $y = 2x^4 - x^2 = x^2(2x^2 - 1)$

The curve is symmetrical about of axis of y.

The curve passes through the origin and the tangent at the origin is $y = 0$, i.e. x-axis

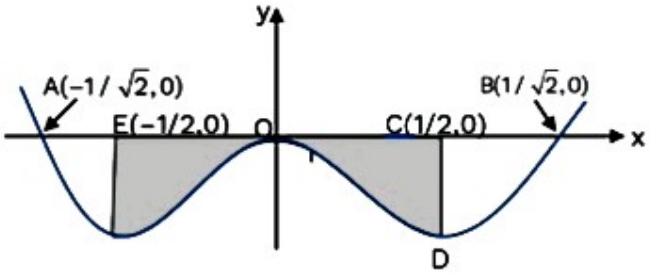
The turning points of the curve are given by

$$\frac{dy}{dx} = 8x^3 - 2x = 0 \Rightarrow 2x(4x^2 - 1) = 0$$

$\Rightarrow$ $x = 0, \pm 1/2$

Now, $\frac{d^2y}{dx^2} = 24x^2 - 2$

Obviously, $\frac{d^2y}{dx^2}$ is +ve when $x = \pm\frac{1}{2}$



and −ve when $x = 0$

$\Rightarrow$ minima at $x = -1/2$ and $x = ½$

At $x = -1/2$, min $y = -1/8$

The curve intersects the axes at O(0,0), $A\left(-\frac{1}{\sqrt{2}}, 0\right)$ and $B\left(\frac{1}{\sqrt{2}}, 0\right)$

Thus, the graph of the curve is known in the figure

Here, $y \le 0$, as x varies from $x = -1/2$ to $x = ½$

$\therefore$ The required area = 2 Area OCDO

$$= 2\left|\int_0^{1/2} y\,dx\right| = 2\left|\int_0^{1/2} (2x^4 - x^2)dx\right| = \frac{7}{120} \text{ sq. units}$$

Q. **The area bounded by the curve $a^2y = x^2(x + a)$ and x-axis is**

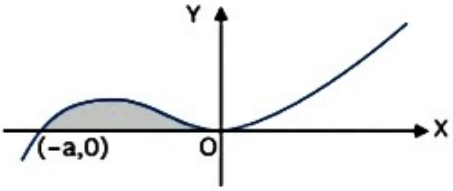
(A) $\frac{a^2}{3}$ **sq. units** **(B)** $\frac{a^2}{4}$ **sq. units**

(C) $\frac{3a^2}{4}$ **sq. units** **(D)** $\frac{a^2}{12}$ **sq. units**

Sol. **(D)**

The curve is $y = \frac{x^2(x + a)}{a^2}$, which is a cubic polynomial.

Since $\frac{x^2(x+a)}{a^2} = 0$ has repeated root x = 0



It touches x-axis at (0,0) and intersects at (–a, 0)

Required area = $\int_{-a}^{0} y\,dx = \int_{-a}^{0}\left[\frac{x^2(x+a)}{a^2}\right]dx = \frac{a^2}{12}$ sq. units

Q. **The area of the loop of the curve, $ay^2 = x^2(a-x)$ is**

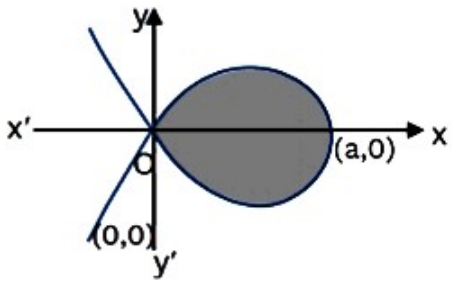
(A) $4a^2$ sq. units **(B) $\frac{8a^2}{15}$ sq. units**

(C) $\frac{16a^2}{9}$ sq. units **(D) None of these**

Sol. **(B)**

$ay^2 = x^2(a-x) \Rightarrow y = \pm x\sqrt{\frac{a-x}{a}}$

Curve tracing: $y = x\sqrt{\frac{a-x}{a}}$



We must have $x \le a$

For $0 < x \le a$, $y > 0$ and for $x < 0$, $y < 0$

Also, $y = 0 \Rightarrow x = 0, a$

Curve is symmetrical about x-axis

When $x \to -\infty$, $y \to -\infty$

Also, it can be verified that y has only one point of maxima for $0 < x < a$.

Area = $2\int_0^a x\sqrt{\frac{a-x}{a}}dx$

$\sqrt{\frac{a-x}{a}} = t \Rightarrow 1 - \frac{x}{a} = t^2 \Rightarrow x = a(1-t^2)$

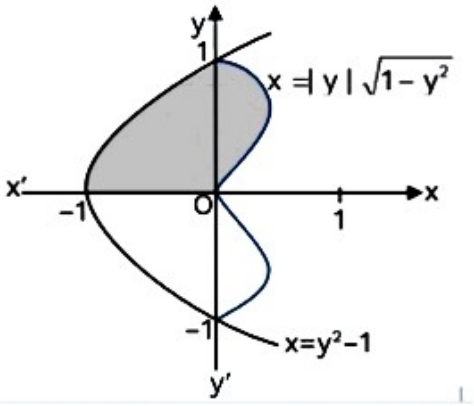
$\Rightarrow \quad A = 2\int_1^0 a(1-t^2)t(-2at)dt$

$= 4a^2\int_0^1 (t^2 - t^4)dt = 4a^2\left[\frac{t^3}{3} - \frac{t^5}{5}\right]_0^1 = 4a^2\left[\frac{1}{3} - \frac{1}{5}\right] = \frac{8a^2}{15}$ sq. units

Q. The area of the region enclosed between the curves $x = y^2 - 1$ and $x = -|y|\sqrt{1-y^2}$ is

(A) 1sq. units **(B)** $\frac{4}{3}$ sq. units **(C)** $\frac{2}{3}$ sq. units **(D)** 2 sq. units

Sol. **(D)**



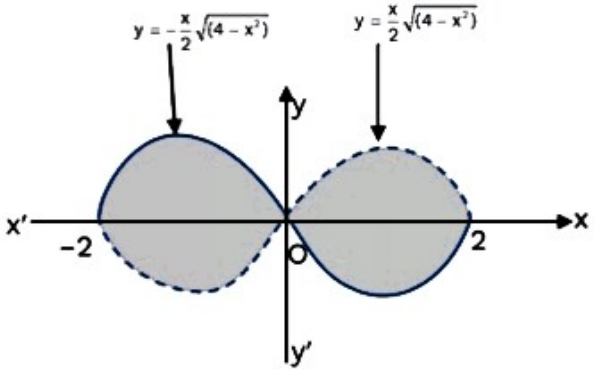
$$A = 2\int_0^1 \left[y\sqrt{1-y^2} - (y^2-1)\right]dy$$

$= 2$ sq. units

Q. The area bounded by the loop of the curve $4y^2 = x^2(4-x^2)$ is

(A) $\frac{7}{4}$ sq. units **(B)** $\frac{8}{3}$ sq. units **(C)** $\frac{11}{3}$ sq. units **(D)** $\frac{16}{3}$ sq. units

Sol. **(D)**



$4y^2 = x^2(4-x^2)$

$\Rightarrow y = \pm\frac{1}{2}\sqrt{x^2(4-x^2)}$

$\Rightarrow y = \pm\frac{x}{2}\sqrt{(4-x^2)}$

$\therefore$ Area(A) $= 4\int_0^2 \frac{x}{2}\sqrt{(4-x^2)}dx$

Let $4 - x^2 = t \Rightarrow -2x\,dx = dt$

$\Rightarrow A = \frac{-4}{4}\int_4^0 \sqrt{t}dt = \int_0^4 \sqrt{t}dt = \left[\frac{t^{3/2}}{3/2}\right]_0^4 = \frac{2}{3}\times\left[\sqrt{64} - 0\right]$

$\Rightarrow A = \frac{16}{3}$ sq. units

Q. **The area enclosed by the curves, $xy^2 = a^2(a - x)$ and $(a - x)y^2 = a^2x$ is**

(A) $(\pi-2)a^2$ sq. units **(B) $(4-\pi)a^2$ sq. units**

(C) $\pi\frac{a^2}{3}$ sq. units **(D) None of these**

Sol. **(A)**

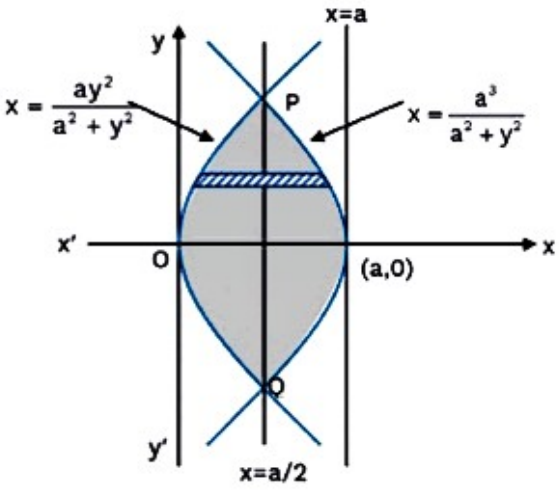
The two curves are

$$xy^2 = a^2(a - x) \Rightarrow x = \frac{a^3}{a^2 + y^2} \quad ...(1)$$

and $(a - x)y^2 = a^2x$

$$\Rightarrow x = \frac{ay^2}{a^2 + y^2} = \frac{ay^2 + a^3 - a^3}{a^2 + y^2} = a - \frac{a^3}{a^2 + y^2} \quad ...(2)$$

Curve (1) is symmetrical about x-axis, and have y-axis as the asymptote.
Curve (2) is symmetrical about x-axis, tangent at origin as y-axis and the asymptote x= a
The two curves intersect at the point P(a/2, a) and Q(a/2, −a)
Required area



$$= 2\int_0^a \left(\frac{a^3}{a^2 + y^2} - \frac{ay^2}{a^2 + y^2}\right)dy$$

$$= 2a\int_0^a \frac{a^2 - y^2}{a^2 + y^2}dy = 2a\left[2\int_0^a \frac{a^2}{a^2 + y^2}dy - \int_0^a dy\right]$$

$$= 2a\left[2a\tan^{-1}\left(\frac{y}{a}\right)_0^a - a\right] = 2a\left[2a\frac{\pi}{4} - a\right] = a^2(\pi - 2)$$

Q. **The area bounded by the curves $y = xe^x$, $y = xe^{-x}$ and the line $x = 1$ is**

(A) $\frac{2}{e}$ sq. units **(B) $1-\frac{2}{e}$ sq. units** **(C) $\frac{1}{e}$ sq. units** **(D) $1-\frac{1}{e}$ sq. units**

Sol. **(A)**

Curve tracing: $y = xe^x$

Let $\frac{dy}{dx} = 0 \Rightarrow e^x + xe^x = 0 \Rightarrow x = -1$

Also, at $x = -1$

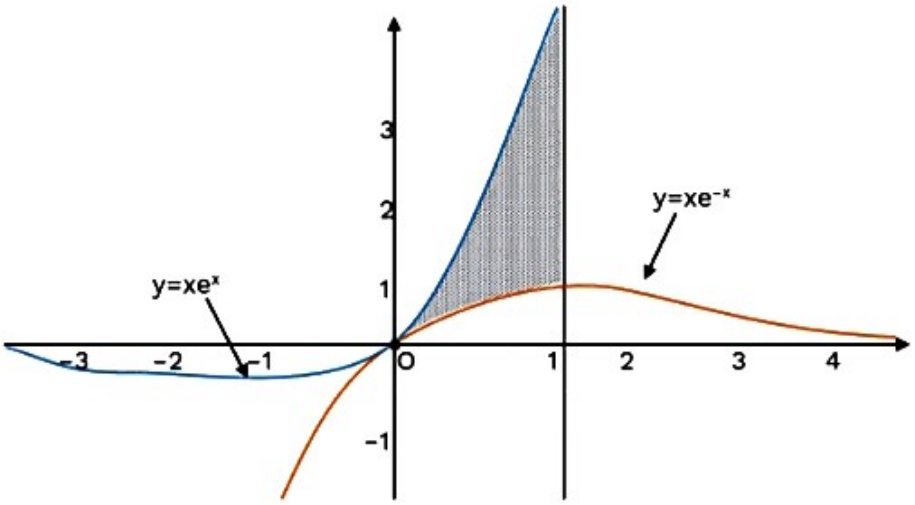
$\frac{dy}{dx}$ changes sign from –ve to +ve

Hence, x = –1 is a point of minima

When $x \to \infty$, $y \to \infty$

Also $\lim\limits_{x\to-\infty} xe^x = \lim\limits_{x\to-\infty} \frac{x}{e^{-x}} = \lim\limits_{x\to-\infty} \frac{(-1)}{e^{-x}} = 0$

With similar types of arguments, we can draw the graph of $y = xe^{-x}$



Required area $= \int_0^1 xe^x dx - \int_0^1 xe^{-x} dx = \left[xe^x\right]_0^1 - \int_0^1 e^x dx - \left(\left[-xe^{-x}\right]_0^1 + \int_0^1 e^{-x} dx\right)$

$= e - (e-1) - \left(-e^{-1} - \left(e^{-1} - 1\right)\right) = \frac{2}{e}$ sq. units.

Q. **The area bounded by the two branches of curve $(y-x)^2=x^3$ and the straight line x=1 is**

(A) $\frac{1}{5}$ **sq. units** **(B)** $\frac{3}{5}$ **sq. units** **(C)** $\frac{4}{5}$ **sq. units** **(D)** $\frac{8}{4}$ **sq. units**

Sol. **(C)**

$(y-x)^2 = x^3$, where $x \geq 0$

$\Rightarrow y - x = \pm x^{3/2}$

$\Rightarrow y = x + x^{3/2}$ ...(1)

$\Rightarrow y = x - x^{3/2}$ ...(2)

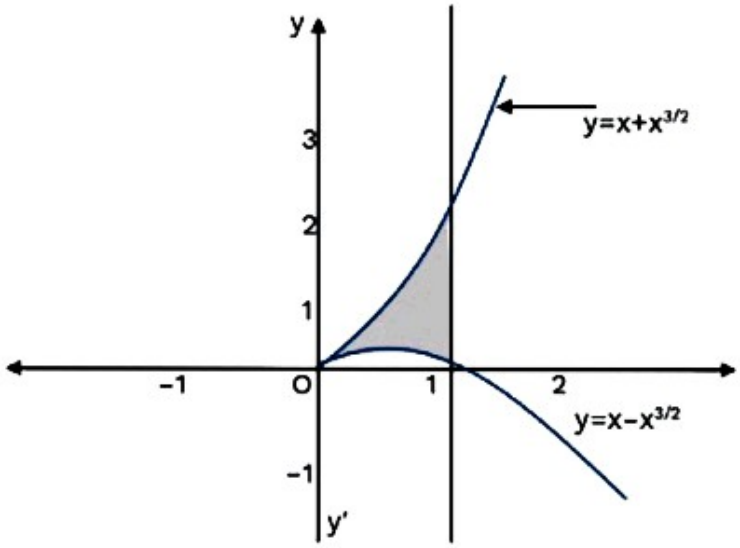
Function (1) is an increasing function.

Function (2) meets x-axis, where $x - x^{3/2} = 0$ or x = 0, 1

Also, for $0 < x < 1$, $x - x^{3/2} > 0$ and for $x > 1$, $x - x^{3/2} < 0$.

When $x \to \infty$, $y \to -\infty$

From these information, we can plot the graph as below:



$$\text{Required area} = \int_0^1 \left[\left(x + x^{3/2}\right) - \left(x - x^{3/2}\right)\right] dx$$

$$= 2\int_0^1 x^{3/2} dx = 2\left[\frac{x^{5/2}}{5/2}\right]_0^1 = \frac{4}{5} \text{ sq. units}$$

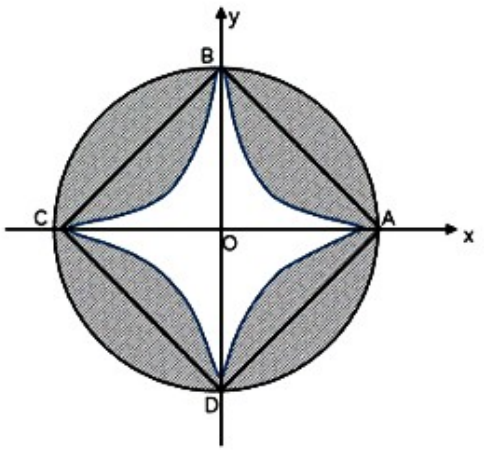
Q. **(a) Sketch and find the area bounded by the curve $\sqrt{|x|} + \sqrt{|y|} = \sqrt{a}$ and $x^2 + y^2 = a^2$ (where $a > 0$).**
(b) If curve $|x| + |y| = a$ divides the area in two parts, then find their ratio in first quadrant only.

Sol. $\sqrt{|x|} + \sqrt{|y|} = \sqrt{a}$

$x = 0 \Rightarrow y = \pm a$

$y = 0 \Rightarrow x = \pm a$

(a) Required area



$$= 4\int_0^a \sqrt{a^2 - x^2} dx - 4\int_0^a \left(\sqrt{a} - \sqrt{x}\right)^2 dx$$

$$= \pi a^2 - 4\int_0^a \left(\sqrt{a} - \sqrt{x}\right)^2 dx$$

$$= \pi a^2 - 4\int_0^a \left[a + x - 2\sqrt{a}\sqrt{x}\right]dx = \pi a^2 - 4\left[a^2 + \frac{a^2}{2} - 2\sqrt{a}\,\frac{2}{3}a^{3/2}\right]$$

$$= \pi a^2 - 4\left[\frac{3a^2}{2} - \frac{4}{3}a^2\right] = \pi a^2 - 4\frac{a^2}{6} = \left(\pi - \frac{2}{3}\right)a^2 \text{ sq. units.}$$

(b) Area included between curves and circle in 1st quadrant

$$= \frac{1}{4}\pi a^2 - \frac{1}{2}a^2 = \frac{(\pi - 2)a^2}{4}$$

Area included between $|x| + |y| = a$ and curve $\sqrt{|x|} + \sqrt{|y|} = \sqrt{a}$ in 1st quadrant

$$\frac{1}{4}\left(\pi - \frac{2}{3}\right)a^2 - \left(\frac{\pi}{4} - \frac{1}{2}\right)a^2 = \frac{a^2}{3}$$

$$\text{Area ratio} = \frac{4}{3(\pi - 2)}$$

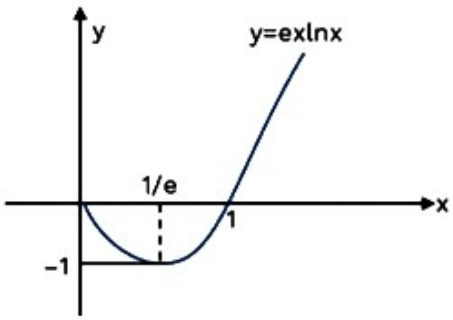
Q. **Area enclosed between the curves $y = ex \cdot \ln x$ and $y = \frac{\ln x}{ex}$**

Sol. $y = ex \ln x$

$$\frac{dy}{dx} = e(1 + \ln x) \qquad \therefore \frac{dy}{dx} = 0 \Rightarrow x = e^{-1}$$

$$\frac{d^2y}{dx^2} = \frac{e}{x} \qquad \therefore \left.\frac{d^2y}{dx^2}\right|_{x=\frac{1}{e}} > 0$$

$\Rightarrow$ minimum at $x = \frac{1}{e}$

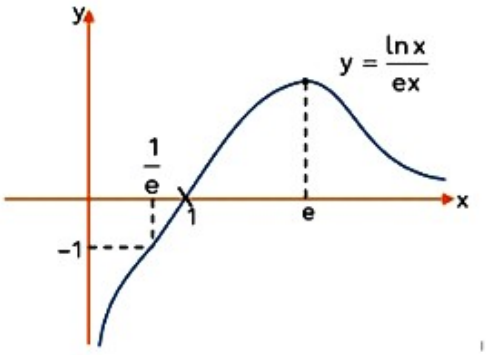


$$y = \frac{\ln x}{ex}$$

$$\frac{dy}{dx} = \frac{1}{e}\left(\frac{1 - \ln x}{x^2}\right)$$

$\therefore$ $\frac{dy}{dx} = 0$ when $x = e$

$$\left.\frac{dy}{dx}\right|_{x=e^+} < 0, \left.\frac{dy}{dx}\right|_{x=e^-} > 0$$



$\Rightarrow$ at $x = e$, $y = \frac{\ln x}{ex}$ has local maxima

$$\text{Required area} = \int_{1/e}^{1}\left(\frac{\ln x}{ex} - ex\ln x\right)dx$$

$$= \left[\frac{(\ln x)^2}{2e} - e\left(\frac{x^2}{2}\ln x - \frac{x^2}{4}\right)\right]_{\frac{1}{e}}^{1} = \frac{e}{4} - \left[\frac{1}{2e} - e\left(\frac{-1}{2e^2} - \frac{1}{4e^2}\right)\right]$$

$$= \frac{e}{4} - \left[\frac{1}{2e} + \frac{3}{4e}\right] = \frac{e}{4} - \frac{5}{4e} = \left(\frac{e^2 - 5}{4e}\right)$$

Q. **Area enclosed by the curve $(y - \sin^{-1}x)^2 = x - x^2$**

Sol. $(y - \sin^{-1}x)^2 = x - x^2$

$y = \sin^{-1}x \pm \sqrt{x - x^2} \Rightarrow$ domain $x \in [0, 1]$

Area enclosed by the curve

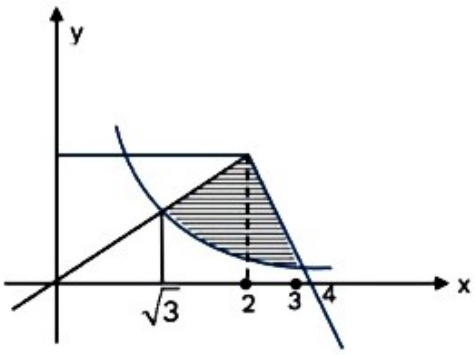
$$= \int_0^1 \left(\sin^{-1} + \sqrt{x - x^2}\right) - \left(\sin^{-1}x - \sqrt{x - x^2}\right)dx = 2\int_0^1 \sqrt{x - x^2}\,dx$$

$$= 2\int_0^1 \sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2} = 2\left[\frac{1}{2}\left(x - \frac{1}{2}\right)\sqrt{x - x^2} + \frac{1}{2}\left(\frac{1}{4}\right)\sin^{-1}\left(\frac{x - \frac{1}{2}}{\frac{1}{2}}\right)\right]_0^1 = \frac{\pi}{4}$$

Q. **Area of the closed figure bounded by the curves $y = 2 - |2 - x|$ and $y = \dfrac{3}{|x|}$**

Sol. $y_1 = 2 - |2 - x| = \begin{cases} x; & x \le 2 \\ 4 - x; & x > 2 \end{cases}$

and $y_2 = \begin{cases} \frac{3}{x}; & x > 0 \\ -\frac{3}{x}; & x < 0 \end{cases}$



so $\frac{3}{x} = x \Rightarrow x = \sqrt{3}$

and $\frac{3}{x} = 4 - x \quad \Rightarrow 3 = 4x - x^2 \Rightarrow x^2 - 4x + 4 = 1$

$\Rightarrow \quad x - 2 = \pm 1 \Rightarrow x = 3$

so required area $= \int_{\sqrt{3}}^{2}\left(x - \frac{3}{x}\right)dx + \int_2^3\left(4 - x - \frac{3}{x}\right)dx$

$$= \left[\frac{x^2}{2} - 3\ln x\right]_{\sqrt{3}}^{2} + \left[4x - \frac{x^2}{2} - 3\ln x\right]_2^3$$

$$= 2 - 3\ln 2 - \frac{3}{2} + 3\ln\left(\sqrt{3}\right) + \left(12 - \frac{9}{2} - 3\ln 3\right) - \left(8 - \frac{4}{2} - 3\ln 2\right)$$

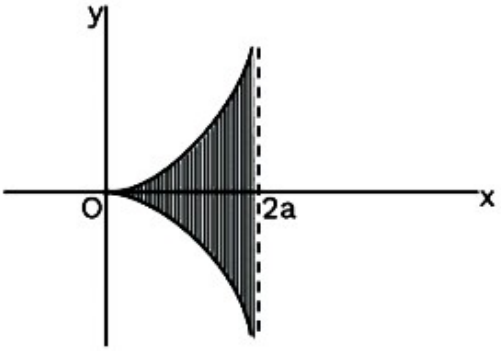
$$= 2 - 3\ln 2 - \frac{3}{2} + \frac{3}{2}\ln 3 + 4 - \frac{5}{2} - 3\ln 3 + 3\ln 2$$

$$= \frac{1}{2} + \frac{3}{2} - \frac{3}{2}\ln 3 = \frac{4 - 3\ln 3}{2}$$

Q. **Find the area enclosed by curve $y^2(2a - x) = x^3$ and its asymptote. $(a > 0)$**

Sol. $y^2 = \dfrac{x^3}{2a - x} \geq 0$

$\Rightarrow x \in [0, 2a)$



Graph

From above diagram, it is clear that asymptote is x = 2a.

By symmetry, we can say that

$$\text{Area enclosed} = 2\int_0^{2a}\left(\frac{x^3}{2a - x}\right)^{\frac{1}{2}} dx$$

$$\Rightarrow A = 2\int_0^{2a} x\sqrt{\frac{x}{2a - x}}\,dx$$

Let $x = 2a\sin^2\theta$

$$\Rightarrow dx = 2a\left(2\sin\theta\cos\theta\right)d\theta$$

$$\Rightarrow A = 2\int_0^{\frac{\pi}{2}} 2a\sin^2\theta\left(\tan\theta\right)2a\left(2\sin\theta\cos\theta\right)d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} 8a^2\left(\sin\theta\right)^4 d\theta$$

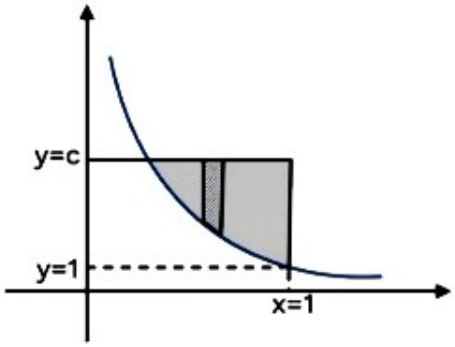
$$= 16a^2\left(\frac{3\pi}{16}\right)$$

$$= 3a^2\pi$$

DETERMINATION OF PARAMETERS:

Q. **Find the value of c for which the area of the figure bounded by the curves $y = \frac{4}{x^2}; x = 1$ and $y = c$ is equal to $\frac{9}{4}$.**

Sol. So required area $= \int_{2/\sqrt{c}}^{1} \left(c - \frac{4}{x^2}\right) dx = \left[cx + \frac{4}{x}\right]_{2/\sqrt{c}}^{1}$



$$\text{Area} = c\left(1 - \frac{2}{\sqrt{c}}\right) + 4 - 2\sqrt{c} = c - 4\sqrt{c} + 4 = \frac{9}{4}$$

$$\Rightarrow \left(\sqrt{c} - 2\right)^2 = \frac{9}{4} \Rightarrow \sqrt{c} = 2 \pm \frac{3}{2}$$

$$\sqrt{c} = \frac{1}{2}, \frac{7}{2}$$

$$c = \frac{1}{4}, \frac{49}{4}.$$

Q. **If the area bounded by $x^2 + 2x = 3$ and the line $y = kx + 1$ is the least, find k and also the least area.**

Sol. x_1 and x_2 are the roots of the equation.

$$x^2 + 2x - 3 = kx + 1, \text{ or}$$

$$x^2 + (2 - k)x - 4 = 0$$

$$\Rightarrow \left.\begin{aligned} x_1 + x_2 &= k - 2 \\ x_1 x_2 &= -4 \end{aligned}\right\}$$

$$A = \int_{x_1}^{x_2} \left[\left(kx + 1\right) - \left(x^2 + 2x - 3\right)\right] dx$$

$$= \left[(k - 2)\frac{x^2}{2} - \frac{x^3}{3} + 4x\right]_{x_1}^{x_2} = \left[(k - 2)\frac{x_2^2 - x_1^2}{2} - \frac{1}{3}\left(x_2^3 - x_1^3\right) + 4(x_2 - x_1)\right]$$

$$= (x_2 - x_1)\left[\frac{(k - 2)^2}{2} - \frac{1}{3}\left(\left(x_2 + x_1\right)^2 - x_1 x_2\right) + 4\right]$$

$$= \sqrt{(k - 2)^2 + 16}\left[\frac{1}{6}(k - 2)^2 + \frac{8}{3}\right]$$

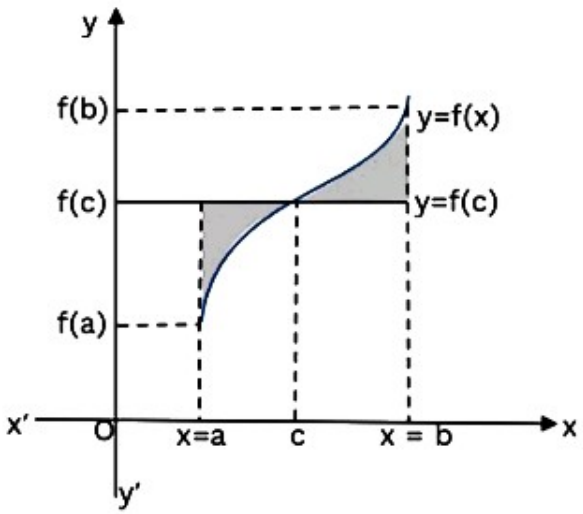
$$= \frac{\left[(k-2)^2 + 16\right]^{3/2}}{6}$$

which is least when k = 2 and A_{least} = 32/3 sq. units.

VARIABLE AREA GREATEST AND LEAST VALUE:

An important concept:

If y = f(x) is a monotonic function in (a, b) then the area bounded by the ordinates at x = a, x = b, y = f(x) and y = f(c), [where $c \in (a, b)$] is minimum when $c = \frac{a+b}{2}$.



Proof :

$$A = \int_a^c (f(c) - f(x))dx + \int_c^b (f(x) - f(c))dx$$

$$= f(c)(c-a) - \int_a^c f(x)dx + \int_c^b (f(x))dx - f(c)(b-c)$$

$$A = \left[2c - (a+b)\right] f(c) + \int_c^b (f(x))dx - \int_a^c (f(x))dx$$

Differentiating w.r.t c,

$$\frac{dA}{dc} = \left[2c - (a+b)\right] f'(c) + 2f(c) + 0 - f(c) - (f(c))$$

for maximas and minima $\frac{dA}{dc} = 0$

$\Rightarrow f'(c)\left[2c - (a+b)\right] = 0$

hence $c = \frac{a+b}{2}$ (as $f'(c) \neq 0$)

Also $c < \frac{a+b}{2}, \frac{dA}{dc} < 0$ and $c > \frac{a+b}{2}, \frac{dA}{dc} > 0$

Hence A is minimum when $c = \frac{a+b}{2}$

Q. **If the area bounded by $f(x) = \frac{x^3}{3} - x^2 + a$ and the straight lines x = 0; x = 2 and the x-axis is minimum then find the value of 'a'.**

Sol. $f(x) = \frac{x^3}{3} - x^2 + a$

$f'(x) = x^2 - 2x = x(x-2) < 0$ (note that f(x) is monotonic in (0, 2))

Hence for the minimum and f(x) must cross the x-axis at $\frac{0+2}{2} = 1$

Hence $f(1) = \frac{1}{3} - 1 + a = 0$

$\Rightarrow \quad a = \frac{2}{3}$

Q. **The value of the parameter a for which the area of the figure bounded by the abscissa axis, the graph of the function $y = x^3 + 3x^2 + x + a$ and the straight lines, which are parallel to the axis of ordinates and cut the abscissa axis at the point of extremum of the function, is the least is**
(A) 2 **(B) 0** **(C) –1** **(D) 1**

Sol. **(C)**

$f(x) = x^3 + 3x^2 + x + a$

$f'(x) = 3x^2 + 6x + 1 = 0$

$\Rightarrow \quad x = -1 \pm \frac{\sqrt{6}}{3}$

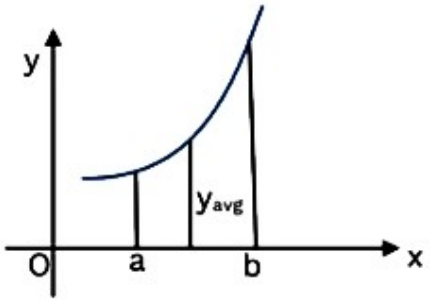
Hence, f(x) cuts the x-axis at $\frac{1}{2}\left[\left(-1+\frac{\sqrt{6}}{3}\right)+\left(-1-\frac{\sqrt{6}}{3}\right)\right] = -1$

$f(-1) = -1 + 3 - 1 + a = 0$
$a = -1.$

AVERAGE VALUE OF A FUNCTION :

Average value of the function in y = f(x) w.r.t x over an interval $a \le x \le b$ is defined as

$$y_{avg} = \frac{1}{b-a}\int_a^b f(x)dx$$



Note :

(i) Average value can be +ve, –ve or zero.

(ii) If the function is defined in (0, ∞) then

$$y_{avg} = \lim_{b\to\infty} \frac{1}{b}\int_a^b f(x)dx \text{ provided the limit exists.}$$

Root mean square value (RMS) is defined as

$$\rho = \left[\frac{1}{b-a}\int_a^b f^2(x)dx\right]^{\frac{1}{2}}$$

Q. **Compute the average value of $f(x) = \dfrac{\cos^2 x}{\sin^2 x + 4\cos^2 x}$ in $\left[0, \dfrac{\pi}{2}\right]$**

Sol. Average value of $f(x) = \dfrac{\cos^2 x}{\sin^2 x + 4\cos^2 x}$

$$y_{average} = \frac{1}{\left(\frac{\pi}{2} - 0\right)}\int_0^{\pi/2} \frac{\cos^2 x}{(\sin^2 x + 4\cos^2 x)}dx = \frac{2}{\pi}\int_0^{\pi/2} \frac{1}{(\tan^2 x + 4)}dx$$

$$= \frac{2}{\pi}\int_0^{\pi/2} \frac{\sec^2 x}{\sec^2 x(\tan^2 x + 4)}dx = \frac{2}{\pi}\int_0^{\pi/2} \frac{\sec^2 x dx}{(1+\tan^2 x)(4+\tan^2 x)}$$

Put $t = \tan x$ so $dt = \sec^2 x\, dx$

$$y_{average} = \frac{2}{\pi}\int_0^{\infty} \frac{dt}{(t^2+1)(4+t^2)} = \frac{2}{3\pi}\int_0^{\infty}\left[\frac{1}{t^2+1} - \frac{1}{t^2+4}\right]dt$$

$$= \frac{2}{3\pi}\left[\tan^{-1} t - \frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right)\right]_0^{\infty} = \frac{2}{3\pi}\left[\frac{\pi}{2} - \frac{\pi}{4}\right] = \frac{2}{3\pi}\frac{\pi}{4} = \frac{1}{6}$$

DETERMINATION OF FUNCTION :

The area function A(x) satisfies the differential equation $\dfrac{dA(x)}{dx} = f(x)$ with initial condition A(a) = 0 i.e. derivative of the area function is the function itself.

Note :

If F(x) is any integral of f(x) then,

$A(x) = \int f(x)dx = [F(x) + c]$

$A(a) = 0 = F(a) + c \Rightarrow c = -F(a)$
hence $A(x) = F(x) - F(a)$.
Finally by taking $x = b$ we get, $A(b) = F(b) - F(a)$

Q. **The area from 0 to x under a certain graph is given to be $A = \sqrt{1+3x} - 1, x \geq 0$:**
(a) Find the average rate of change of A w.r.t x as x increases from 1 to 8
(b) Find the instantaneous rate of change of A w.r.t x at x = 5
(c) Find the ordinate (height) y of the graph as a function of x.
(d) Find the average value of the ordinate (height) y, w.r.t x as x increases from 1 to 8

Sol. $A = \sqrt{1+3x} - 1 = \int_0^x f(x)dx$

(a) $\left.\frac{dA}{dx}\right|_{avg} = \frac{1}{(8-1)}\int_1^8 \left(\frac{dA}{dx}\right)dx$

$= \frac{1}{7}\left(\sqrt{1+3x} - 1\right)_1^8 = \frac{1}{7}(4-1) = \frac{3}{7}$

(b) $\left.\frac{dA}{dx}\right|_{x=5} = \frac{3}{2\sqrt{1+3x}} = \frac{3}{2\sqrt{1+3(5)}} = \frac{3}{8}$

(c) $A(x) = \int_0^x f(x)dx = \sqrt{1+3x} - 1$

Differentiating w.r.t x

$f(x) = \frac{3}{2\sqrt{1+3x}}$

(d) $y_{avg} = \frac{1}{(8-1)}\int_1^8 f(x)dx = \frac{1}{7}\left(\sqrt{1+3x}\right)_1^8 = \frac{3}{7}$

Q. **Let C_1 and C_2 be the graphs of the functions $y = x^2$ and $y = 2x$, $0 \leq x \leq 1$ respectively. Let C_3 be the graph of a function $y=f(x)$, $0 \leq x \leq 1$, $f(0) = 0$. For a point P on C_1 let the lines through P, parallel to the axes, meet C_2 and C_3 at Q and R respectively (see figure). If for every position of P (on C_1), the areas of the shaded regions OPQ & ORP are equal, determine the function f(x).**

Sol. Let $P(h, h^2)$ be a point on the curve C_1
$\Rightarrow$ R(h, f(h))
Area OPQO = Area OPRO

$$\int_0^{h^2}\left(\sqrt{y}-\frac{y}{2}\right)dy=\int_0^{h}\left(x^2-f(x)\right)dx$$

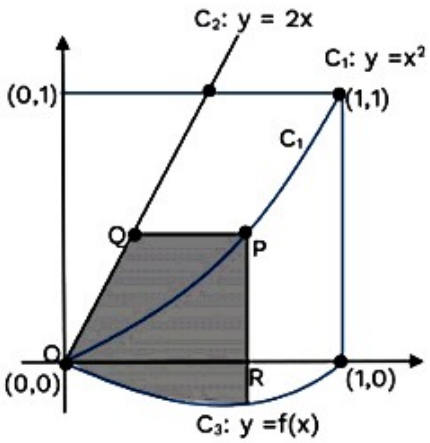
Differentiating w.r.t h

$$\left(\sqrt{h^2}-\frac{h^2}{2}\right)\cdot 2h = h^2 - f(h)$$

$$\Rightarrow 2h^2 - h^3 = h^2 - f(h)$$

$$\Rightarrow f(h) = h^3 - h^2$$

$$\Rightarrow f(x) = x^3 - x^2$$



AREA ENCLOSED IN CASE ONE CURVE IS EXPRESSED IN POLAR FORM :

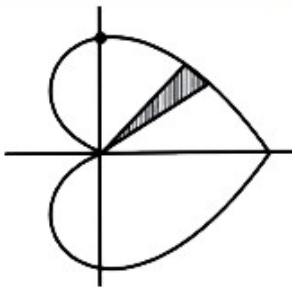
$$\text{Area of any curve} = \frac{1}{2}\int r^2 d\theta$$

Q. **Find the area of the cardioid r = a(1 + cos θ)**

Sol.

$$A = \frac{1}{2}\int_0^{2\pi} r^2 d\theta = \frac{a^2}{2}\int_0^{2\pi} 4\cos^4\frac{\theta}{2}d\theta \quad \text{put } \frac{\theta}{2} = t$$

$$A = a^2\int_0^{\pi} 4\cos^4 t\,dt = 8\times\frac{3\pi a^2}{16} = \left(\frac{3\pi a^2}{2}\right)$$



AREA IN RESPECT OF CURVE REPRESENTED PARAMETRICALLY:

Q. **Find the area enclosed by the curves x = a sin³t and y = acos³t**

Sol.

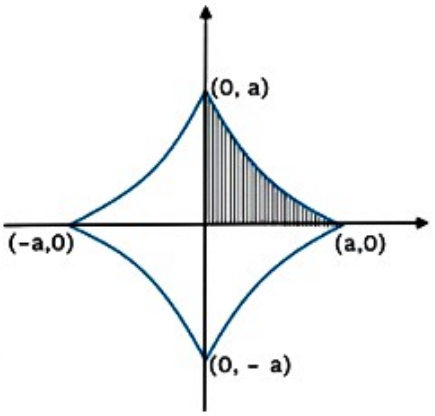
$$x^{2/3} + y^{2/3} = a^{2/3}$$

$$\text{Required area } 4\int_0^{a}\left(a^{2/3} - x^{2/3}\right)^{3/2}dx$$

Put $x = a\sin^3 t$; $dx = 3a\sin^2 t\cos t\,dt$

$$\text{Area} = 4\int_0^{\pi/2}\left(a^{2/3} - a^{2/3}\sin^2 t\right)^{3/2} 3a\sin^2 t\cos t\,dt$$

$$A = 12a^2\int_0^{\pi/2}\sin^2 t\cos^4 t\,dt \qquad \ldots(i)$$



$$A = 12a^2 \int_0^{\pi/2} \sin^4 t \cos^2 t dt \qquad \text{...(ii)}$$

Adding (i) and (ii)

$$A = \frac{12a^2}{2} \int_0^{\pi/2} \cos^2 t \sin^2 t(\cos^2 t + \sin^2 t)dt$$

$$= 6a^2 \int_0^{\pi/2} \sin^2 t \cos^2 t dt = 6a^2 \int_0^{\pi/2} \frac{\sin^2 2t}{4} = \frac{3a^2}{2} \int_0^{\pi/2} \left(\frac{1 - \cos 4t}{2} \right) dt$$

$$= \frac{3a^2}{4} \left(t - \frac{\sin 4t}{4} \right)_0^{\pi/2} = \frac{3a^2}{4} \left(\frac{\pi}{2} \right) = \frac{3\pi a^2}{8}$$